
Arsenic & the Reported Cancers

Ladera Ranch / South Orange County - Scientific Evidence Repository

Physician-directed | attorney-supervised | hypothesis-neutral

Compiled evidence base, protocols, and phased gated plan

Version 2026-07-23 | no causation asserted | PHI excluded

This compiles the investigation's public narrative deliverables. Every substantive claim carries an evidence level (a established fact ... g unknown) and a citation. The private patient registry (folder 04) is excluded from this and every shared output. This is not a legal conclusion and does not establish that any party caused any illness.

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00 PROJECT GOVERNANCE

Project Scope

PROJECT SCOPE - Arsenic & the Ladera Ranch reported cancers

Status: foundation established 2026-07-23. This is a physician-directed, attorney-supervised evidence-repository build. It follows the completed historical investigation (mandated 1907-1912 arsenical cattle dipping in South Orange County) and does not re-litigate that history except to verify a medical or exposure claim.

Core research question

Could historical inorganic arsenic exposure plausibly cause, initiate, promote, accelerate, or interact with genetic susceptibility in the development of Ewing sarcoma and the other confirmed rare cancers reported among Ladera Ranch residents?

"Arsenic and cancer" is not sufficient. The build focuses on cancer specificity, pediatric exposure, tissue of origin, molecular mechanism, exposure timing, and dose.

What this repository is / is not

- Is: a transparent, reproducible, source-graded evidence base so independent physicians, epidemiologists, toxicologists, molecular biologists, environmental scientists, and attorneys can judge exactly what is established, plausible, unresolved, or unsupported.
- Is not: an advocacy document, a causation claim, or a legal conclusion.

Evidence grading (applied to every substantive statement)

`a. established fact | b. strong evidence | c. suggestive evidence | d. biological plausibility | e. hypothesis | f. inference | g. unknown`

Non-negotiable rules

- Follow the evidence; never confirm a predetermined conclusion.
- Do not state arsenic causes Ewing sarcoma unless a source states that exact conclusion.
- Every substantive claim carries an inline citation + a full source record (see DATABASE_SCHEMA).
- Prefer primary research, government reports, systematic reviews, authoritative monographs.
- Never fabricate a citation, DOI, PMID, quotation, statistic, author, study, or conclusion.
- A documented absence of research is a finding - never converted into evidence of no effect.
- Capture contradictory/negative evidence with equal diligence.
- Flag retractions, expressions of concern, preprints, and predatory journals.
- PHI stays private (folder 04, gitignored). Public outputs use aggregation/masking only.
- Do not contact families, witnesses, agencies, researchers, or defendants. No legal conclusions.
- Preserve originals + hashes; never overwrite source files.

Priority order (active)

1. Direct arsenic-Ewing literature, incl. negative findings and gaps. (in progress)
2. Arsenic mechanisms that could create structural DNA damage / gene fusions / defective repair / selection of fusion-positive cells.
3. Ewing etiology, environmental epidemiology, developmental biology.
4. Disease-specific arsenic evidence for every other verified cancer.
5. Historical-exposure biomarkers + tumor-testing opportunities at risk of loss.
6. Experiments testing EWSR1 breakage / fusion / repair / selection under environmentally relevant As.
7. CDC/ATSDR-compliant epidemiological + spatial design.

Reproducibility standard

Independent experts must be able to re-run every search (SEARCH_LOG.md), inspect every source

(02_source_library + hashes), and challenge every assumption (19_claims_audit). The output is judged by reproducibility and defensibility, not persuasiveness.

00 PROJECT GOVERNANCE

Phased Gated Execution Plan

PHASED & GATED EXECUTION PLAN

Arsenic & the reported cancers - physician-directed, attorney-supervised
Version 1.0 | 2026-07-23 | hypothesis-neutral | no causation asserted

This plan sequences all 16 workstreams into 6 phases separated by decision gates. It defines what proceeds automatically at full authority versus what must stop at a human gate. It is designed so independent experts can see exactly where the project is, what has been cleared, and what is blocked.

1. AUTHORITY MODEL ("full authority," scoped)

Granted - Research Engine executes autonomously, at full authority, continuously:
desk literature search across all named databases; source capture + grading + hashing; evidence / mechanism / claims matrices; synthesis docs; drafting protocols, analysis plans, briefs, request letters, and checklists; QA and reproducibility logging; committing/pushing to the repo **with the private folder (04) excluded**.

Reserved - never opens without the named gate owner's written authorization (these are also the standing prohibitions of this project):

Reserved action | Gate owner

Collecting/handling any biological specimen (teeth, nails, blood spots, FFPE) | Physician lead + Ethics/IRB
Soil/dust/water sampling on any property | Attorney + landowner/agency permission
Any handling of PHI / case-identifying data (folder 04) | Physician lead + Ethics/IRB
Contacting families, witnesses, agencies, researchers, defendants | Attorney (project does not contact)
Legal conclusions or advocacy language | Attorney (out of scope for this engine)
Public release of any output | Physician lead + Attorney
Initiating lab/tumor testing | Physician lead + Ethics/IRB

Standing rules that never gate open: no fabricated citations; absence-of-research recorded as a finding; contradictory evidence weighted equally; source-before-significance; PHI stays private; preserve originals + hashes. (See PROJECT_SCOPE.)

2. PHASE OVERVIEW

Phase | Name | Nature | Authority | Exit gate

0 | Governance & scaffold | desk | auto | - (complete)
1 | Desk evidence base | desk | auto (full authority) | GATE A - scientific integrity review
2 | Designs & protocols | desk | auto (drafting) | GATE B - expert + ethics + legal sign-off
3 | Evidence preservation | real-world prep->action | draft auto / act gated | GATE C - holds executed
4 | Acquisition & testing | real-world | gated (experts/labs) | GATE D - results QA'd
5 | Integration & reporting | desk->gated release | auto draft / gated release | GATE E - release decision

3. PHASES IN DETAIL

PHASE 0 - Governance & scaffold [x] COMPLETE

Deliverables done: 24-folder repo; PROJECT_SCOPE, SEARCH_PROTOCOL, DATABASE_SCHEMA; SEARCH_LOG; seeded matrices; privacy guard (04 gitignored).

PHASE 1 - Desk evidence base [~] IN PROGRESS (~55%) | authority: AUTO

Objective: exhaust the accessible literature; grade and cross-check every claim.

Workstreams: 1(taxonomy, public fields only) | 2 mechanisms | 3 direct As-Ewing | 4 bridge | 5 other cancers | 6 cattle-dip chemistry | 7 environmental fate | 8 biomarkers | 9 molecular signatures | 13 alternatives.

Done: 3,4,5,8 first pass; 10 design (Phase-2 item, pulled early); GIS schema.

Remaining: native-database re-runs (PubMed/Europe PMC/OpenAlex/etc.) with logged counts; 6, 7, 9, 13; complete matrices; dual-review consequential findings; CONTRADICTIONARY_EVIDENCE.md; TOP_100 sources; TOP_25 gaps; TOP_20 hypotheses; TOP_15 falsifiers.

Exit criteria -> GATE A: every substantive claim cited + graded; searches reproducible; dual-review done; contradictory evidence catalogued; no unresolved fabrication/integrity flags.

GATE A - Scientific integrity review. Owner: independent scientific reviewers (toxicology, molecular oncology, environmental epidemiology). Criteria: reproducibility of SEARCH_LOG; source integrity; balanced capture of negative evidence; correct evidence levels. Output: sign-off or revise list. Blocks Phase 2 protocolization until passed.

PHASE 2 - Designs & protocols [o] NOT STARTED | authority: AUTO (draft only; no execution)

Objective: convert findings into testable, review-ready protocols. No real-world action.

Deliverables: 14 critical-experiments dossier (design/controls/power/endpoints/blinding/ethics); blinded tumor molecular + metallomics protocol (9); biomarker exposure-reconstruction protocol (8/10); authorized soil-sampling SAP (analyte set: total+bioaccessible As, lead, speciation; 0-6 in); preregistration of the epi plan (12); GIS build spec (11); 12 causal-inference weight-of-evidence framework (Bradford Hill + source-pathway-receptor + mode-of-action + AOP).

Exit criteria -> GATE B: protocols complete, powered, ethically specified, and internally consistent with Phase-1 evidence.

GATE B - Expert + Ethics/IRB + Legal review. Owners: domain experts; Ethics/IRB (human subjects, specimens, PHI); Attorney (chain of custody, discovery, privilege). **This is the gate that can authorize real-world action.** Output: approved protocols + explicit authorizations (which specimens/sites/analyses, under what custody). Nothing physical happens before GATE B.

PHASE 3 - Evidence preservation [o] NOT STARTED | authority: DRAFT auto / ACT gated

Objective: stop loss of destroyable evidence (time-critical).

Draft now (auto): IMMEDIATE_EVIDENCE_PRESERVATION_CHECKLIST [x]; draft legal-hold letters, records requests, specimen-retention guidance.

Act (gated, human): attorney issues legal holds (newborn blood-spot cards, FFPE blocks, ranch/county/state archives); families retain keepsakes via their own clinicians/counsel; soil at priority grounds secured before any grading/remediation under authorized chain of custody.

Exit -> GATE C: holds in place; priority soil access permitted.

GATE C - Preservation confirmed. Owner: Attorney + Physician lead. Criteria: holds executed; custody established. Unblocks acquisition.

PHASE 4 - Acquisition & testing [o] NOT STARTED | authority: GATED (experts/labs execute)

Objective: generate the missing measurements. **Executed by authorized clinicians/labs, not this engine;** the engine supports design fidelity, data management, and blinded analysis interpretation.

Activities (each pre-authorized at GATE B): soil sampling + speciation; biomarker reconstruction (teeth/nails/blood spots); blinded tumor WGS/long-read/RNA-seq/methylation + metallomics; CCR-based epidemiology per the preregistered plan; GIS integration (masked public / private analytic).

Exit -> GATE D: results quality-controlled, chain of custody intact, analyses per preregistration.

GATE D - Results integrity. Owner: scientific reviewers + biostatistician. Criteria: QA/QC, no preregistration deviations unlogged, PHI protected. Unblocks integration.

PHASE 5 - Integration & reporting [o] NOT STARTED | authority: AUTO draft / GATED release

Objective: weigh all evidence; state exactly what is established / plausible / unresolved / excluded.

Deliverables: updated 12 causal weight-of-evidence; final STUDY_EVIDENCE_MATRIX; updated DOCTOR_TECHNICAL_BRIEF and ATTORNEY_SCIENCE_BRIEF; EXECUTIVE_SUMMARY; general vs specific causation statement; research-gaps for any unresolved links.

Exit -> GATE E: briefs internally consistent, dual-reviewed, PHI-clean.

GATE E - Release decision. Owners: Physician lead + Attorney. Criteria: accuracy, privacy, appropriate hedging, no advocacy overreach. Decides what (if anything) is shared, with whom, and whether public materials in 23_..._hold are ever released.

4. DEPENDENCIES (critical path)

Phase 1 -> GATE A -> Phase 2 -> GATE B (authorizes reality) -> Phase 3 preservation -> Phase 4 testing -> GATE D -> Phase 5 -> GATE E. Preservation (3) runs in parallel and urgently once GATE B clears the relevant specimens/sites, because those items degrade.

5. RISK REGISTER (top)

Risk | Impact | Mitigation

Specimens/soil lost before holds | irreversible | Phase 3 urgency; draft holds now; flag at every gate
 Texas-Sharpshooter / cluster overreach | credibility, legal | preregistration (12); verified counts only
 PHI leak to public repo | ethical/legal | 04 gitignored; pre-commit check every push
 Over-strong causal language | Daubert/credibility | evidence levels a-g on every claim; dual review
 Fabricated/incorrect citation | fatal to defensibility | never-fabricate rule; native-db verification at GATE A
 Confounder neglect (orchard lead-arsenate, DDT, ancestry) | biased inference | Workstream 13 + ancestry standardization

6. PROGRESS TRACKER

- Phase 0 [x] | Phase 1 [x] (all WS first pass + native-db done; dual-review = GATE A, human)
- Phase 2 [~] DRAFTS complete: soil SAP (E1), biomarker protocol (E2), blinded tumor protocol (E3), epi preregistration form, draft preservation/records letters - all pending GATE A review + GATE B authorization
- Gates A-E: not yet reached.

7. IMMEDIATE NEXT (auto, Phase 1) - proceeding under full authority

1. Workstream 12 causal-inference weight-of-evidence framework (Bradford Hill etc.).
2. Workstream 6 cattle-dip chemistry crosswalk (As trioxide vs arsenite/arsenate; CAS; soil fate).
3. Workstream 7 environmental fate & bioavailability.
4. Native-database confirmation of Priority-1/2 citations; begin CONTRADICTORY_EVIDENCE.md.
5. Assemble GATE-A package (TOP_100 sources, TOP_25 gaps, TOP_20 hypotheses, TOP_15 falsifiers).

Executive Summary

EXECUTIVE SUMMARY - Arsenic & the reported cancers (foundation established 2026-07-23)

Where this stands

The evidence-repository **framework is built and the Priority-1 question has a first, honest, sourced answer.** This is the start of a large multi-session scientific build, not a finished review. Nothing here asserts causation.

The single most important finding so far (Priority 1)

- No study links inorganic arsenic to Ewing sarcoma. Ewing has *no established environmental risk factors* and is defined by the EWSR1::FLI1 fusion. -> documented gap, not a negative result.
- **Arsenic is a Group 1 human carcinogen - but for bladder, lung, and skin (epithelial cancers), not bone/mesenchymal sarcoma.** Evidence for one tumor type does not transfer to another.
- A general genotoxic bridge (oxidative DNA damage, DNA-repair inhibition, genomic instability) is biologically plausible but has never been connected to the specific EWSR1 fusion.
- Full detail + citations: 05_ewing_sarcoma/DIRECT_ARSENIC_EWING_FINDINGS.md.

The pivotal unknown (drives everything downstream)

Could environmentally relevant inorganic arsenic **increase EWSR1-region DNA breaks, impair their repair, or select fusion-positive progenitor cells?** No one has tested this. It is the crux of WORKSTREAM 4 and the proposed critical experiments.

What is in place (this pass)

- 24-folder reproducible repository; PHI folder (04) gitignored - never enters the public repo.
- Governance: PROJECT_SCOPE, SEARCH_PROTOCOL, DATABASE_SCHEMA.
- SEARCH_LOG with the two executed searches + the queued native-database strings.
- Seeded matrices (bibliography, evidence, claims, Ewing-mechanism, other-cancers, biomarkers, experts).
- Priority-1 findings with 8 real sources (IARC 100C, As-carcinogenesis review PMID 30223072, Ewing GWAS, NCI PDQ, etc.), each flagged for dual-review + full-record capture.

Honest limitations of this pass

- Searches used a web index as a doorway into primary databases; **native-database re-runs (PubMed/Europe PMC/etc.) with logged counts and full source records are pending.**
- Single-reviewer. Consequential findings are flagged for dual verification.
- Workstreams 2-16 are scaffolded but not yet populated.

Standard of success

Independent experts can reproduce every search, inspect every source, and see exactly what is established (a), plausible (d), or unknown (g). Persuasiveness is not the metric; reproducibility is.

Gate A Readiness

GATE A READINESS (scientific-integrity / dual-review)

Phase 1 desk evidence base is ~90% assembled and ready to package for GATE A review.

Ready for review

- Findings: 05 direct As-Ewing; 07 mechanistic bridge; 06 other cancers; 08 dip chemistry; 09 molecular signatures; 10 epi plan; 11 GIS; 12 causal weight-of-evidence; 13 alternatives; 14 proposed experiments.
- Synthesis: DOCTOR + ATTORNEY briefs; EXECUTIVE_SUMMARY.
- Integrity aids: CONTRADICTORY_EVIDENCE; TOP_25 gaps; TOP_20 hypotheses; TOP_15 falsifiers.
- Matrices: bibliography (29 sources), evidence/mechanism/other-cancer/biomarker/claims; SEARCH_LOG (19 searches).

Still to complete before formally passing GATE A

1. Native-database re-runs (PubMed/EuropePMC/OpenAlex) of all Workstream-3 strings with logged counts + full source records + sha256 (currently web-index-sourced, flagged).
2. Dual-reviewer verification of consequential findings (currently single-reviewer 'RE').
3. Expand bibliography toward TOP_100; capture any papers found in native re-runs.

Gate A owners / criteria (per PHASED_GATED_EXECUTION_PLAN)

Owners: independent toxicology + molecular-oncology + environmental-epidemiology reviewers.

Criteria: reproducibility of SEARCH_LOG; source integrity (no fabrication); balanced negative-evidence capture; correct a-g evidence levels. Output: sign-off or revise-list. Blocks Phase 2 until passed.

Headline for reviewers

Arsenic->Ewing causation is currently unsupported/unestablished (weak Bradford-Hill, MOA mismatch, tissue-tropism incoherence, counter-analogy), while the **environmental exposure predicate is plausible but unmeasured locally.** The defensible path is measurement (soil speciation + biomarker reconstruction + blinded tumor analysis + preregistered epidemiology), not assertion.

Search Protocol

SEARCH PROTOCOL

1. For each research question, run the exact string in each listed database; log id/date/db/string/filters/count/exclusions in SEARCH_LOG.md.
2. Databases (WORKSTREAM order): PubMed, PMC, Europe PMC, Crossref, OpenAlex, Semantic Scholar, Google Scholar, Scopus/WoS (if access), ProQuest, univ. repositories; agencies IARC, NTP, ATSDR, EPA/IRIS, NIEHS, NCI, WHO, OEHHA, DTSC, USGS, USDA, CDPH, California Cancer Registry, ClinicalTrials.gov.
3. Secondary articles are used only to locate primary sources; never cited as the evidence.
4. Every retained hit -> full source record (DATABASE_SCHEMA) + sha256 of the PDF/HTML where possible.
5. Flag retractions/EoC/preprints/predatory venues. Capture negative + contradictory results equally.
6. Absence of results is recorded as a finding, with the exact strings that returned nothing.
7. Dual-review the consequential items before they enter any physician/attorney brief.

Database Schema

DATABASE SCHEMA (source + evidence records)

MASTER_BIBLIOGRAPHY.csv / SOURCE_INTEGRITY_LOG.csv (one row per source)

source_id, title, authors, year, journal_or_publisher, doi, pmid, pmcid, url, access_date,
source_type(primary_research|review|systematic_review|monograph|govt_report|registry|preprint|
dissertation|conference|secondary), peer_reviewed(y/n), grade(A1|A2|B1|B2|C|D), retraction_flag,
expression_of_concern, predatory_flag, funding_conflicts, full_text_available, file_path, sha256,
notes, reviewer, verification_status(unverified|single|dual)

STUDY_EVIDENCE_MATRIX.csv (one row per claim/study)

evidence_id, disease, exposure, arsenic_species, exposure_route, exposure_window, study_type,
population_or_model, sample_size, outcome, effect_estimate, ci, significance, dose_response,
mechanism, supports(y/n), contradicts(y/n), neutral(y/n), directness(direct|indirect|analogous),
relevance(high|med|low), risk_of_bias, key_limitations, replication_status, citation, doi, pmid,
url, access_date, full_text, notes, reviewer, verification_status

CLAIMS_AUDIT.csv (one row per assertion)

claim_id, claim_text, domain, source_ids, classification(verified|partially|plausible_unverified|
unsupported|contradicted|needs_expert|needs_env_testing|needs_epi|needs_medical_records|needs_discovery),
evidence_level(a-g), notes, reviewer

Mechanism/biomarker/other-cancer matrices follow PROJECT spec fields; headers seeded in 18_evidence_matrices.

Cancer Registry Template

CANCER TAXONOMY TEMPLATE (Workstream 1) - schema only; DATA lives PRIVATELY in 04 (gitignored)

Physician/legal teams populate one record per diagnosis in 04_cancer_registry_private (never committed).
Public outputs use aggregation/masking only. Do not place identifiers here.

Fields per case:

standardized_disease_name; histologic_subtype; primary_tumor_location; age_at_diagnosis; sex;
year_of_diagnosis; year_began_residence_or_school; residence_exposure_window; school_exposure_window;
prenatal_parental_residence_window; known_genetic_predisposition; prior_therapeutic_radiation_or_chemo;
pathology_confirmation_status; molecular_findings; EWSR1_fusion_status; EWSR1_fusion_partner;
bone_or_extrasketal; ever_lived_outside_area; other_major_exposures; data_provenance;
verification_status.

Rules: separate research branch per confirmed cancer; do not combine biologically unrelated cancers;
diagnoses verified against the California Cancer Registry (not social media). PHI never leaves 04.

Final Integrated Report Skeleton

FINAL INTEGRATED REPORT - SKELETON (Phase 5 template, populate after testing)

Status: empty template. Populated only after GATE D (results QA'd). Release only at GATE E (physician + attorney). Every populated claim carries an evidence level (a-g) + citation. Hypothesis-neutral; states what is established / plausible / unresolved / excluded. PHI aggregated/masked.

0. Abstract / plain-language summary

[what was asked | what was measured | what was found | what remains unknown]

1. Scope & question

Pull from PROJECT_SCOPE. Restate general vs specific causation separation.

2. Methods & reproducibility

Search protocol + SEARCH_LOG counts; DATABASE_SCHEMA; grading a-g; dual-review record; preregistration ID/date-lock (epi); protocols executed (E1-E3) + deviations log; chain of custody.

3. The historical predicate (verify-only, from prior investigation)

Mandated dipping; arsenical chemistry (Workstream 6); documented Joplin ground; what remains inference. Cite the historical repo; do not re-argue it.

4. Environmental results

- Soil (E1): total + bioaccessible As, lead, speciation by location/depth vs background (~12 mg/kg). Dip signature (low-lead As trioxide) vs orchard (As+Pb). [results | level]
- Environmental fate / redistribution during grading (Workstream 7). [results | level]

5. Exposure results (individual + population)

- Biomarkers (E2): reconstructed As timeline (teeth/nails/blood spots), etiologic window, affected vs comparison; methylation signature. Temporality + dose. [results | level]
- GIS integration (Workstream 11), masked. [results | level]

6. Disease-pattern results (epidemiology)

Per preregistered plan: SIR + exact CI, Bayesian, spatial scan, sensitivity + negative controls; ancestry-standardized; verified CCR counts only. State per decision rules, not bare p-values. [results | level]

7. Biological results (tumor)

- Blinded tumor (E3): tumor arsenic (metallomics), arsenic methylation signature, EWSR1 breakpoint architecture, SV burden - with low-mutation-burden caveat. [results | level]
- Mechanistic experiments (Workstream 4, F-series) if run: EWSR1-region breakage/repair/selection.

8. Alternatives & confounders (Workstream 13)

Lead-arsenate (orchard), DDT/organochlorines, ancestry, chance, ascertainment; how each was addressed.

9. Causal weight-of-evidence

Integrate via CAUSAL_WEIGHT_OF_EVIDENCE (Bradford Hill + source-pathway-receptor + MOA/AOP + triangulation). General causation verdict. Then, only if supported, specific-causation map per individual (private). Use GENERAL_VS_SPECIFIC_CAUSATION structure.

10. What is established / plausible / unresolved / excluded

Four explicit lists with levels + citations.

11. Falsification outcomes

Which F-series findings occurred; what they weaken/strengthen (either direction).

12. Limitations

Power, specimen scarcity, signature non-specificity, historical inference, residual confounding.

13. Research still required

From TOP_25 gaps; what would move remaining g/d cells to a/b.

14. Briefs

Link/append DOCTOR_TECHNICAL_BRIEF + ATTORNEY_SCIENCE_BRIEF (final versions).

15. Appendices

MASTER_BIBLIOGRAPHY + hashes; STUDY_EVIDENCE_MATRIX; CLAIMS_AUDIT; SEARCH_LOG; protocols; preregistration; QA/QC; deviation logs. (Private registry stays in 04, not in any released version.)

Do not release without GATE E. A "clean/null" outcome is a legitimate, reportable result and must be stated as plainly as a positive one.

01 SEARCH LOGS

Search Log

SEARCH LOG (reproducible)

Each entry: id | date | database/engine | exact string | filters | result count | action.

Note: 2026-07-23 searches used a general web engine as an index into primary databases; each hit must be re-run in the native database (PubMed/Europe PMC/etc.) and captured with full fields before use in a brief. Native-database re-runs are pending (see NEXT_30_PRIORITY_ACTIONS).

id | date | engine | query | results | action

SL-2026-07-23-01 | 2026-07-23 | web index | arsenic Ewing sarcoma EWSR1 fusion inorganic arsenic exposure study | 8 shown | 0 direct As-Ewing studies; Ewing = FET::ETS fusions, "no established environmental risk factors" -> Finding 1

SL-2026-07-23-02 | 2026-07-23 | web index | IARC arsenic group 1 carcinogen established cancer sites lung skin bladder genotoxicity mechanism | 17 shown | Group 1; sufficient human evidence = bladder/lung/skin (epithelial); genotoxic + DNA-repair-inhibition mechanisms -> Findings 2 & 3

Native-database strings queued (WORKSTREAM 3 variants) - to run + log

arsenic AND "Ewing sarcoma"; "inorganic arsenic" AND Ewing; "arsenic trioxide" AND Ewing;
 arsenite AND Ewing; arsenate AND Ewing; arsenic AND EWSR1; arsenic AND "EWS-FLI1";
 arsenic AND "ETS translocation"; arsenic AND "fusion oncogene"; arsenic AND "pediatric bone cancer";
 arsenic AND "small round blue cell tumor"; arsenic AND "mesenchymal stem cell" transformation;
 arsenic AND "DNA double-strand breaks"; arsenic AND "nonhomologous end joining";
 arsenic AND "homologous recombination"; arsenic AND chromothripsis; arsenic AND "genomic instability".

Databases: PubMed, PMC, Europe PMC, Crossref, OpenAlex, Semantic Scholar, Scholar, ProQuest, NTP, IARC, ATSDR, EPA IRIS, OEHHA, ClinicalTrials.gov. Record filters, counts, exclusions each run.

2026-07-23 session 2 (mechanism, etiology, other cancers, biomarkers)

id | query | results | action

SL-2026-07-23-03 | arsenic DNA double-strand breaks NHEJ HR repair inhibition | 6 | As2O3 lowers HR not NHEJ (PMID 37530740) -> Bridge A

SL-2026-07-23-04 | Ewing cell of origin MSC EWSR1-FLI1 fusion timing prenatal | 8 | embryonic MSC origin; fusion impairs repair (PMC2952797; Nat Commun 2025)

SL-2026-07-23-05 | Ewing environmental risk agricultural pesticide parental occupation | 7 | farm/pesticide case-control signal (PMID 12020108;1311140;15551353)

SL-2026-07-23-06 | arsenic chromosomal translocation gene fusion carcinogenesis | 8 | error-prone repair -> translocations; As2O3 treats APL (PMC5421966)

SL-2026-07-23-07 | arsenic osteosarcoma/bone/soft-tissue sarcoma | 9 | STS (angiosarcoma est.; vineyard STS); osteosarcoma no direct

SL-2026-07-23-08 | arsenic leukemia/lymphoma/kidney/liver | 10 | kidney/liver suggestive; leuk/lymph inconsistent (PMC9099091)

SL-2026-07-23-09 | deciduous teeth arsenic reconstruction biomarker | 6 | tooth-ring metal biomarker validated (s41370-021-00400-x)

SL-2026-07-23-10 | newborn blood spots / toenail / hair arsenic biomarker | 6 | toenail in-utero As (jes201438); banked DBS (Clin Chem 2026)

2026-07-23 session 3 (epidemiology / spatial)

id | query | results | action

SL-11 | CDC ATSDR cancer cluster guidelines 2022 criteria | 10 | 2022 guidelines; cluster def; 10-criteria; de-emphasize significance (S016)

SL-12 | Ewing incidence per million SEER age-specific | 9 | ~2.93-3/M <20; peak 10-15; M:F 3:1; 9x white (S017)

SL-13 | SaTScan spatial scan Poisson Bayesian SIR | 8 | SaTScan methods; multi-scale scans (S020)

SL-14 | cluster pitfalls Texas sharpshooter multiple comparisons | 8 | silent multiplicity; pre-specify boundaries; source-before-test (S018,S019)

2026-07-23 session 4 (causal MOA / dip chemistry / fate)

id | query | results | action

SL-15 | arsenic MOA AOP key characteristics IARC | 8 | dominant MOA cytotoxicity/regeneration, oft non-genotoxic (S021,S022) -> tempers plausibility

SL-16 | As2O3/arsenite/arsenate dip formulation CAS toxicity | 8 | arsenite from As2O3+NaOH; arsenite 10x arsenate; sulfhydryl MOA (S026)

SL-17 | former cattle-dip vat soil arsenic contamination Australia | 8 | up to ~3000 mg/kg As + DDT; persists decades; residential exceedances (S023,S024,S025)

2026-07-23 session 5 (molecular signatures)

id | query | results | action

SL-18 | arsenic mutational/methylation signature tumor WGS | 8 | As lung T>G signature (S027); 255-CpG methylation (S028)

SL-19 | Ewing genomic landscape low mutation burden STAG2 | 7 | 0.15 mut/Mb -> low signature power (S029)

2026-07-23 session 6 (NATIVE DB re-runs, reproducible counts)

Ran 12 Workstream-3 strings vs PubMed E-utilities + Europe PMC REST (see NATIVE_DB_SEARCH_RESULTS.md).

Key: arsenic+"Ewing sarcoma" PubMed=8 (ALL therapeutic ATO), arsenic+EWSR1=0, arsenic+EWS-FLI1=0, "inorganic arsenic"+Ewing=0. Characterized the 8 hits by title (all ATO-as-therapy).

2026-07-23 session 7 (TOP-100 native capture)

Captured 76 real PubMed records across 20 topical strings (relevance-sorted) -> 02_source_library/

TOP_100_MOST_RELEVANT_SOURCES.csv. Notable new: arsenite impairs BRCA1-dependent DSB repair (37762697); BRCA2-HR needed for arsenite lesions (19553191) - strengthen Bridge A with specific citations.

01 SEARCH LOGS

Native Db Search Results

NATIVE-DATABASE SEARCH RESULTS (reproducible hit counts)

Date 2026-07-23. APIs: NCBI E-utilities (PubMed) + Europe PMC REST. Counts are live hitCounts; re-runnable by anyone with the same strings. Direct arsenic-Ewing strings return ~0 -> confirms the documented absence (Finding 1, 05_) with reproducible evidence.

Query | PubMed | EuropePMC | Top IDs (PubMed)

arsenic AND "Ewing sarcoma" | 8 | 323 | 31746397, 27665785, 22391311, 22315235, 21183792
 "inorganic arsenic" AND Ewing | 0 | 28 | -
 "arsenic trioxide" AND "Ewing sarcoma" | 8 | 207 | 31746397, 27665785, 22391311, 22315235, 21183792
 arsenite AND Ewing | 3 | 105 | 41738287, 33876544, 8264344
 arsenate AND Ewing | 15 | 35 | 42213593, 39671892, 31746397, 27665785, 22391311
 arsenic AND EWSR1 | 0 | 147 | -
 arsenic AND "EWS-FLI1" | 0 | 76 | -
 arsenic AND "primitive neuroectodermal tumor" | 1 | 69 | 16930595
 arsenic AND "small round blue cell tumor" | 0 | 11 | -
 arsenic AND "mesenchymal stem cell" AND transformation | 0 | 226 | -
 arsenic AND "double-strand break" | 23 | 627 | 39032852, 38096740, 37762697, 37530740, 36089002
 arsenic AND translocation AND sarcoma | 10 | 1211 | 37601428, 29346731, 29307831, 27058871, 26283888

Interpretation: low/zero counts on the direct As-Ewing strings are a reproducible documented absence of research, not evidence of no relationship. Mechanistic/analog strings (double-strand break, MSC transformation, translocation-sarcoma) return the mechanism literature already captured in 07_/14_.

CRITICAL CHARACTERIZATION - the arsenic-Ewing hits are THERAPEUTIC, not etiologic

All 8 PubMed hits for arsenic AND "Ewing sarcoma" (= the same 8 for "arsenic trioxide" AND "Ewing sarcoma") concern arsenic trioxide (ATO) as a candidate TREATMENT of Ewing, not a cause:

- 31746397 (2020) ATO + etoposide interaction in Ewing cell lines
- 27665785 (2016) ATO potentiates etoposide in Ewing sarcomas
- 22315235 (2012) ATO evaluated by the Pediatric Preclinical Testing Program (Ewing focus)
- 21183792 (2011) ATO inhibits cancer growth by blocking Hedgehog signaling
- 22391311 (2012) ATO + Hedgehog signaling review
- 21183780 (2011) "Arsenic: a potentially useful poison for Hedgehog-driven cancers"
- 16889724 (2006) ATO combination chemo (osteosarcoma)
- 16646077 (2006) ATO induces p53-independent Ewing cell death

arsenic AND EWSR1 = 0, arsenic AND EWS-FLI1 = 0, "inorganic arsenic" AND Ewing = 0.

Finding (a, reproducible): the published arsenic-Ewing record is entirely arsenic-as-therapy (ATO kills Ewing cells, akin to its role in APL). Zero etiologic studies (arsenic causing Ewing) exist. This is a documented absence of causal research AND a directional counter-signal.

Annotated Key Sources

ANNOTATED KEY SOURCES (abstracts captured)

Curated pivotal sources underpinning the findings. Abstracts fetched from PubMed (efetch) on 2026-07-23 for reviewer verification. Full source-record + sha256 capture per DATABASE_SCHEMA still required at GATE A. Abstracts are the authors' words (verbatim), not this engine's summary.

PMID 37530740 - MECH: As decreases homologous recombination, not NHEJ

<https://pubmed.ncbi.nlm.nih.gov/37530740/>

1. FEBS J. 2023 Nov;290(22):5313-5321. doi: 10.1111/febs.16922. Epub 2023 Aug 8. Arsenic affects homologous recombination and single-strand annealing but not end-joining pathways during DNA double-strand break repair. Kurosawa A(1)(2)(3), Saito S(1), Sakurai M(1), Shinozuka M(1), Someya Y(2), Adachi N(1). Author information: (1)Graduate School of Nanobioscience, Yokohama City University, Japan. (2)Faculty of Science and Technology, Gunma University, Maebashi, Japan. (3)Gunma University Center for Food and Science and Wellness, Gunma University, Maebashi, Japan. Arsenic is a carcinogen that can cause skin, lung, and bladder cancer. While DNA double-strand breaks (DSBs) have been implicated in arsenic-induced carcinogenesis, the exact mechanism remains unclear. In this study, we performed genetic analysis to examine the impact of arsenic trioxide (As₂O₃) on four different DSB repair pathways using the human pre-B cell line Nalm-6. Random integration analysis showed that As₂O₃ does not negatively affect non-homologous end joining or polymerase theta-mediated end joining. In contrast, ...

PMID 37762697 - MECH: Arsenite impairs BRCA1-dependent DSB repair -> genomic instability

<https://pubmed.ncbi.nlm.nih.gov/37762697/>

1. Int J Mol Sci. 2023 Sep 21;24(18):14395. doi: 10.3390/ijms241814395. Arsenite Impairs BRCA1-Dependent DNA Double-Strand Break Repair, a Mechanism Potentially Contributing to Genomic Instability. Matthäus T(1), Stößer S(1), Seren HY(1), Haberland VMM(1), Hartwig A(1). Author information: (1)Department of Food Chemistry and Toxicology, Institute of Applied Biosciences (IAB), Karlsruhe Institute of Technology (KIT), Adenauerring 20a, 76131 Karlsruhe, Germany. BRCA1 is a key player in maintaining genomic integrity with multiple functions in DNA damage response (DDR) mechanisms. Due to its thiol-rich zinc-complexing domain, the protein may also be a potential target for redox-active and/or thiol-reactive (semi)metal compounds. The latter includes trivalent inorganic arsenic, which is indirectly genotoxic via induction of oxidative stress and inhibition of DNA repair pathways. In the present study, we investigated the effect of NaAsO₂ on the transcriptional and functional DDR. Particular attention was paid to the potential impairment of BRCA1-mediated DDR mechanisms by arsenite by compa ...

PMID 10712745 - MECH: As(III) induces intrachromosomal homologous recombination (hprt)

<https://pubmed.ncbi.nlm.nih.gov/10712745/>

1. Environ Mol Mutagen. 2000;35(2):114-22. doi: 10.1002/(sici)1098-2280(2000)35:2<114::aid-em6>3.0.co;2-q. Arsenic[III] and heavy metal ions induce intrachromosomal homologous recombination in the hprt gene of V79 Chinese hamster cells. Helleday T(1), Nilsson R, Jenssen D. Author information: (1)Department of Genetic and Cellular Toxicology, Wallenberg Laboratory, Stockholm University, Stockholm, Sweden. helleday@genetics.su.se In the present study the carcinogenic metal ions Cd[II], Co[II], Cr[VI], Ni[II], and Pb[II], as well as As[III], were examined for their ability to induce intrachromosomal homologous and nonhomologous recombination in the hprt gene of two V79 Chinese hamster cell lines, SPD8 and Sp5, respectively. With the exception of Pb[II], all of these ions enhanced homologous recombination, the order of potency being Cr>Cd>As>Co>Ni. In contrast, Cr[VI] was the only ion to enhance recombination of the nonhomologous type. In order to obtain additional information on the mechanism of recombination in the SPD8 cell line, individual clones exhibiting metal-induced recombination ...

PMID 30223072 - REVIEW: arsenic carcinogenesis - epidemiology/metabolism/genotoxicity/epigenetics

<https://pubmed.ncbi.nlm.nih.gov/30223072/>

1. Regul Toxicol Pharmacol. 2018 Nov;99:78-88. doi: 10.1016/j.yrtph.2018.09.010. Epub 2018 Sep 15. A review on arsenic carcinogenesis: Epidemiology, metabolism, genotoxicity and epigenetic changes. Zhou Q(1), Xi S(2). Author information: (1)Department of Environmental and Occupational Health, Liaoning Provincial Key Laboratory of Arsenic Biological Effect and Poisoning, School of Public Health

, China Medical University, No.77 Puhe Road, Shenyang North New Area, Shenyang 110122, Liaoning Province, People's Republic of China. (2)Department of Environmental and Occupational Health, Liaoning Provincial Key Laboratory of Arsenic Biological Effect and Poisoning, School of Public Health, China Medical University, No.77 Puhe Road, Shenyang North New Area, Shenyang 110122, Liaoning Province, People's Republic of China. Electronic address: Shxi@cmu.edu.cn. Long-term exposure to arsenic (inorganic arsenic) is a world-wide environmental health concern. Arsenic is classified as the Group 1 human carcinogen by the International Agency for Research on Cancer (IARC). Epidemiological studies have est ...

PMID 32971865 - GENE-ENV: AS3MT and methylation of inorganic arsenic (metabolism/susceptibility)

<https://pubmed.ncbi.nlm.nih.gov/32971865/>

1. Biomolecules. 2020 Sep 22;10(9):1351. doi: 10.3390/biom10091351. Arsenic Methyltransferase and Methylation of Inorganic Arsenic . Roy NK(1), Murphy A(1), Costa M(1). Author information: (1)Department of Environmental Medicine, NYU School of Medicine, 341 East 25th Street, New York, NY 10010, USA. Arsenic occurs naturally in the environment, and exists predominantly as inorganic arsenite (As(III)) and arsenate As(V)). Arsenic contamination of drinking water has long been recognized as a major global health concern. Arsenic exposure causes changes in skin color and lesions, and more severe health conditions such as black foot disease as well as various cancers originating in the lungs, skin, and bladder. In order to efficiently metabolize and excrete arsenic, it is methylated to monomethylarsonic and dimethylarsinic acid. One single enzyme, arsenic methyltransferase (AS3MT) is responsible for generating both metabolites. AS3MT has been purified from several mammalian and nonmammalian species, and its mRNA sequences were determined from amino acid sequences. With the advent of genome ...

PMID 20514360 - EPIGEN: arsenic effects on DNA methylation and epigenetic gene regulation

<https://pubmed.ncbi.nlm.nih.gov/20514360/>

1. Epigenomics. 2010 Feb;2(1):87-104. doi: 10.2217/epi.09.45. Effects of arsenic exposure on DNA methylation and epigenetic gene regulation. Reichard JF(1), Puga A. Author information: (1)Department of Environmental Health & Center for Environmental Genetics, University of Cincinnati College of Medicine, 3223 Eden Avenue, Cincinnati, OH 45267-0056, USA. John.Reichard@uc.edu Arsenic is a nonmutagenic human carcinogen that induces tumors through unknown mechanisms. A growing body of evidence suggests that its carcinogenicity results from epigenetic changes, particularly in DNA methylation. Changes in gene methylation status, mediated by arsenic, have been proposed to activate oncogene expression or silence tumor suppressor genes, leading to long-term changes in the activity of genes controlling cell transformation. Mostly descriptive, and often contradictory, studies have demonstrated that arsenic exposure is associated with both hypo- and hyper-methylation at various genetic loci in vivo or in vitro. This ambiguity has made it difficult to assess whether the changes induced by arsenic ...

PMID 12020108 - EPI: Ewing sarcoma - farm exposures & parental pesticide occupation (Australia case-control)

<https://pubmed.ncbi.nlm.nih.gov/12020108/>

1. Cancer Causes Control. 2002 Apr;13(3):263-70. doi: 10.1023/a:1015036109130. Farm exposures, parental occupation, and risk of Ewing's sarcoma in Australia: a national case-control study. Valery PC(1), McWhirter W, Sleight A, Williams G, Bain C. Author information: (1)Population & Clinical Sciences Division, Queensland Institute of Medical Research, Royal Brisbane Hospital, Australia. patriciV@qimr.edu.au OBJECTIVE: It has been suggested that parental occupation, particularly farming, increased the risk of Ewing's sarcoma in the offspring. In a national case-control study we examined the relationship between farm and other parental occupational exposures and the risk of cancer in the offspring. METHODS: Cases were 106 persons with confirmed Ewing's sarcoma or peripheral primitive neuroectodermal tumor. Population-based controls (344) were selected randomly via telephone. Information was collected by interview (84% face-to-face). RESULTS: We found an excess of case mothers who worked on farms at conception and/or pregnancy (odds ratio (OR)=2.3. 95% confidence interval (CI) 0.5-12.0) a ...

Direct Arsenic Ewing Findings

DIRECT ARSENIC-EWING SARCOMA FINDINGS (Priority 1)

Search date: 2026-07-23 | Reviewer: research-engine (single-reviewer; flag for dual review)
 Bottom line up front: No study directly linking inorganic arsenic exposure to Ewing sarcoma was located. This is a documented absence of research, not evidence of no relationship. Arsenic is an established human carcinogen, but for different tissue types (epithelial: bladder, lung, skin), not bone/mesenchymal sarcomas. A mechanistic bridge exists only at the level of **general genotoxicity / DNA-repair inhibition (biological plausibility, level d) and has not** been connected to the specific EWSR1 translocation that defines Ewing sarcoma.

Evidence levels used: a established fact | b strong | c suggestive | d biological plausibility | e hypothesis | f inference | g unknown.

Finding 1 - No direct arsenic-Ewing literature located | level (a) for the absence

A targeted search (see SEARCH_LOG entry SL-2026-07-23-01) returned zero studies examining inorganic arsenic exposure as a risk factor for Ewing sarcoma. Reviews and GWAS of Ewing etiology describe "no established environmental risk factors"; the disease is defined by FET::ETS fusions (EWSR1::FLI1 in >85%). Arsenic is not discussed as a candidate exposure.

- Grossmann/GWAS susceptibility loci: PMC6085378; 2025 GWAS meta-analysis PMC12919121.
- NCI Ewing Sarcoma Treatment (PDQ), health-professional version - etiology section.

Do not convert this absence into "arsenic does not cause Ewing sarcoma." It means the specific question has never been studied. It defines a research gap (see 20_research_gaps).

Finding 2 - Arsenic is a Group 1 carcinogen, but for epithelial sites, not sarcoma | level (a)

IARC classifies inorganic arsenic as Group 1 (carcinogenic to humans). Sites with **sufficient human evidence: urinary bladder, lung, skin (non-melanoma). These are carcinomas (epithelial)**.
 Bone sarcoma / Ewing (mesenchymal) is not among the established arsenic cancer sites.

- IARC Monograph Vol. 100C, "Arsenic and Arsenic Compounds" - NCBI Bookshelf NBK304380; overall evaluation NBK533643; tumour-site concordance NBK570318.
- Specificity caveat (level a): established evidence for arsenic concerns a different cell lineage than Ewing. Evidence for one tumor type does not transfer to another (governance rule).

Finding 3 - A general genotoxic bridge exists but is not Ewing-specific | level (d)

Arsenic carcinogenesis mechanisms (well characterized for its established sites) include: oxidative DNA damage (ROS), inhibition of DNA repair (base-excision repair; binding zinc-finger repair proteins such as PARP), chromosomal aberrations, genomic instability, cell-cycle disruption, and epigenetic dysregulation.

- Review: Arsenic carcinogenesis - epidemiology, metabolism, genotoxicity, epigenetics. PubMed 30223072 (Sci Total/Regul Toxicol Pharmacol, 2018).
- Arsenic disruption of DNA-damage responses: PMC4693233 (Biomolecules 2015).
- Arsenic co-carcinogenesis / inhibition of DNA repair + zinc-finger proteins: ScienceDirect S1044579X21001371.

The bridge is plausible but non-specific. Inhibited repair + genomic instability is a mechanism by which translocations can arise/persist in general; **no source shows arsenic produces or selects the EWSR1 fusion. Do not equate general DNA damage with proof of an EWSR1-fusion mechanism (governance rule). This is the central question for WORKSTREAM 4.**

Net Priority-1 assessment

Question | Answer | Level

Direct human study, As -> Ewing? | None located | absence (a)

Is arsenic a human carcinogen? | Yes, Group 1 | a

For bone/Ewing/sarcoma specifically? | No sufficient human evidence | a

Plausible genotoxic mechanism that could bear on fusion sarcomas? | Yes, general only | d

Arsenic shown to create/select the EWSR1 fusion? | No evidence located | g / gap

What this defines for the next steps

1. WORKSTREAM 4 mechanistic bridge is the pivotal unknown: can environmentally relevant iAs increase EWSR1-region double-strand breaks, impair their repair, or select fusion-positive progenitors? (See 16_proposed_experiments - critical experiments 12-15.)
2. Molecular-pathological epidemiology: is there a validated *arsenic mutational/methylation signature*, and could it be tested (blinded) against Ewing tumors? (WORKSTREAM 9.)
3. Specificity discipline: every downstream claim must respect that arsenic's proven cancers are epithelial, and that Ewing has a single defining genetic lesion of unknown environmental trigger.

Source records (to be promoted to MASTER_BIBLIOGRAPHY.csv with full fields + access hashes)

- IARC Monograph 100C Arsenic - <https://www.ncbi.nlm.nih.gov/books/NBK304380/> (A1)
- IARC overall evaluation (Group 1) - <https://www.ncbi.nlm.nih.gov/books/NBK533643/> (A1)
- Tumour-site concordance & mechanisms - <https://www.ncbi.nlm.nih.gov/books/NBK570318/> (A1)
- As carcinogenesis review, PubMed 30223072 - <https://pubmed.ncbi.nlm.nih.gov/30223072/> (A1)
- As & DNA-damage responses, PMC4693233 - <https://pmc.ncbi.nlm.nih.gov/articles/PMC4693233/> (A1)
- Ewing GWAS susceptibility, PMC6085378 - <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6085378/> (A1)
- Ewing GWAS meta-analysis 2025, PMC12919121 - <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12919121/> (A1)
- NCI Ewing Sarcoma Treatment (PDQ) - <https://www.cancer.gov/types/bone/hp/ewing-treatment-pdq> (A2)

Single-reviewer pass. Consequential items (Findings 1-3) flagged for dual-reviewer verification and for full source-record capture per DATABASE_SCHEMA before use in any physician/attorney brief.

UPDATE 2026-07-23 - native-database confirmation (decisive)

Native PubMed/Europe PMC re-runs (01_search_logs/NATIVE_DB_SEARCH_RESULTS.md) confirm Finding 1 with reproducible counts AND add a decisive nuance:

- arsenic AND EWSR1 = 0 PubMed; arsenic AND EWS-FLI1 = 0; "inorganic arsenic" AND Ewing = 0.
- The only arsenic + Ewing PubMed hits (n=8) are arsenic trioxide as a candidate THERAPY (kills Ewing cells / blocks Hedgehog), not arsenic as a cause.

Upgraded statement (level a): there is zero etiologic literature connecting arsenic to Ewing, and the sole existing arsenic-Ewing science shows arsenic killing Ewing cells - the opposite direction. This is a documented absence of causal research plus a directional counter-signal; it is still not evidence of "no relationship," but any claim that arsenic causes Ewing has **no literature support whatsoever.**

Other Cancers Arsenic Findings

OTHER REPORTED CANCERS x ARSENIC (Workstream 5)

Search date: 2026-07-23 | single-reviewer (flag for dual review). Per governance, evidence for one cancer does not transfer to another. Classes: established causal | sufficient | limited | suggestive | mechanistic/animal only | no direct evidence | evidence against | not adequately studied.

Cancer | Arsenic relationship (class) | Key evidence | Note

Bladder | Established (IARC sufficient human) | IARC 100C; urinary-tract review PMC4088919 | epithelial; dose-response in drinking water

Lung | Established (IARC sufficient) | IARC 100C | epithelial

Skin (non-melanoma) | Established (IARC sufficient) | IARC 100C | epithelial

Kidney | Suggestive / positive association | IARC "positive association"; PMC4088919 | not "sufficient"

Liver | Suggestive; hepatic angiosarcoma established | STS review (angiosarcoma + arsenic/Thorotrast/vinyl chloride) | angiosarcoma = a soft-tissue sarcoma

Prostate | Suggestive (positive association) | IARC | -

Soft-tissue sarcoma (STS) | Limited/suggestive; occupational | arsenic a listed STS risk factor; STS excess among vineyard workers (arsenical-pesticide exposed) | closest sarcoma signal; mesenchymal

Osteosarcoma (bone) | No direct epidemiological evidence located | sarcoma epidemiology PMC3564705 | pesticide-osteosarcoma signal exists but not a arsenic-specific

Ewing sarcoma (bone) | No direct evidence (see 05_ewing_sarcoma) | - | the primary question; gap

Leukemia / lymphoma | Limited & inconsistent; possible dual effect | Front Public Health 2022 (PMC9099091): As has dual carcinogenic/therapeutic effects | As₂O₃ treats APL

CNS/brain, thyroid | Not adequately studied for As | - | pending WORKSTREAM-5 native search

Signals most relevant to a sarcoma hypothesis

- Hepatic angiosarcoma is an established arsenic-caused sarcoma (mesenchymal/endothelial) - proof arsenic can cause a sarcoma in humans, at a specific site (liver vasculature). Does not extend to bone/Ewing, but defeats any claim that "arsenic only causes carcinomas."
- STS in vineyard workers (arsenical pesticides) - occupational arsenical -> soft-tissue sarcoma. Suggestive, not bone-specific.

Early-life / in-utero (relevant to a pediatric hypothesis)

- In-utero + childhood arsenic -> increased young-adult mortality from bladder/laryngeal/liver cancer (Antofagasta cohort, PMC3556614); in-utero arsenic -> DNA damage/micronuclei in newborns (Environ Health 2019, s12940-019-0481-7). Establishes early-life vulnerability windows - not a pediatric-bone-cancer link.

Honest net

Arsenic's human cancer footprint is real but site-specific and largely epithelial, with **one established sarcoma (hepatic angiosarcoma) and a suggestive occupational STS signal. For bone sarcomas (osteosarcoma, Ewing) there is no direct human evidence** - a gap, not a refutation.

Sources

IARC 100C (NBK304380); urinary-tract As review PMC4088919; leukemia/lymphoma dual-effect PMC9099091; in-utero mortality PMC3556614; in-utero DNA damage s12940-019-0481-7; sarcoma epidemiology PMC3564705.

Workstream4 Mechanistic Bridge

WORKSTREAM 4 - Could arsenic create or favor an EWSR1 fusion? (mechanistic bridge)

Search date: 2026-07-23 | single-reviewer (flag for dual review) | no causation asserted.

Evidence levels: a fact | b strong | c suggestive | d biological plausibility | e hypothesis | f inference | g unknown.

Central question: could environmentally relevant inorganic arsenic increase EWSR1-region double-strand breaks (DSBs), bias their repair toward translocation, or select/expand a fusion-positive mesenchymal progenitor?

Net position: a genuine, plausible bridge now exists - arsenic suppresses high-fidelity DSB repair in ways associated with chromosomal translocations, Ewing's cell of origin is a developmentally restricted mesenchymal progenitor, and Ewing has a real (if unresolved) agricultural/pesticide epidemiological signal. **But no study connects arsenic to the specific EWSR1 fusion, and there is meaningful counter-evidence. The bridge is level d (plausible)**, not demonstrated.

Bridge A - Arsenic suppresses homologous recombination, shifting to error-prone repair linked to translocations | level (d), strengthened

- Arsenic trioxide decreases homologous recombination (HR) and single-strand annealing but does not impair non-homologous end joining (NHEJ), in human pre-B Nalm-6 cells - PMID 37530740.
- Arsenic-altered DSB repair **favors error-prone pathways (NHEJ) and is associated with chromosomal translocations** and genomic instability - "Oncogenomic disruptions in arsenic-induced carcinogenesis," PMC5421966 (Oncotarget).
- Relevance: Ewing is defined by a translocation (t(11;22) EWSR1::FLI1). A carcinogen that demotes error-free HR while leaving error-prone joining intact is a mechanism by which mis-joined breaks (translocations) can arise/persist in general.
- Directness to Ewing: INDIRECT. No data on the EWSR1/FLI1 loci, on DSB frequency at those loci, or in mesenchymal progenitors. -> the critical missing experiment.

Bridge B - Genomic instability / LINE-1 derepression | level (d)

- Arsenic dysregulates LINE-1 retrotransposons, driving genomic instability (PMC5421966). Instability is a permissive backdrop for structural rearrangement. Non-specific to EWSR1.

Bridge C - Epigenetic reprogramming (Polycomb/H3K27me3) | level (d/e)

- Arsenic disrupts H3K9me3/H3K27me3 balance via PRC2 (PALI1 inhibition) - PMC12223765.
- EWS::FLI1 oncogenesis is itself a chromatin-reprogramming disease. Whether arsenic's epigenetic shifts intersect Ewing enhancer biology is untested -> hypothesis.

Bridge D - Developmental susceptibility window | level (d/e)

- Ewing's cell of origin is a **mesenchymal stem/progenitor cell, most permissive at the embryonic stage**; EWS::FLI1 in human embryonic MSCs yields Ewing-like tumors, and the fusion causes transcriptional reprogramming + defective DNA-damage repair - PMC2952797; Nat Commun 2025 s41467-025-64475-y.
- Arsenic crosses the placenta; prenatal/early-life is an established arsenic susceptibility window.
- Convergence (hypothesis): if the fusion arises prenatally/early in a mesenchymal progenitor, the timing of arsenic exposure could matter. No data test this for Ewing.

Cross-cutting epidemiology - Ewing ? agricultural / pesticide exposure | level (c) for pesticides, (g) for arsenic specifically

- Real, replicated signal (contra the flat "no environmental risk factors" summary):
- Australia national case-control: risk doubled with maternal pesticide/insecticide handling; more cases lived on farms - PMID 12020108.
- Paternal agricultural occupation / herbicide-pesticide-fertilizer exposure, RR ~6.1 (SF study); PMIDs 1311140, 15551353.
- Systematic review: significant pesticide-Ewing/osteosarcoma association.
- California perinatal fine-particulate study - PMC11887085 (different exposure, shows this community/registry is actively studied).
- Bridge to Ladera: historical arsenicals (lead-arsenate, arsenic trioxide) were agricultural pesticides / the cattle-dip. But "pesticide" studies did not isolate arsenic - they cannot be read as arsenic-specific evidence. This defines a measurement gap, not proof.

COUNTER-EVIDENCE (weighted equally)

1. Arsenic trioxide treats a translocation cancer. As₂O₃ is standard curative therapy for acute promyelocytic leukemia (APL, t(15;17) PML-RARA fusion) - arsenic can kill fusion-driven cells, not only create risk. Direction of effect is context-dependent.
2. EWS-FLI1 alone does not transform normal human cells; cooperating events are required, and the fusion itself impairs DNA repair (Nat Commun 2025). The rate-limiting step may not be break formation at all.
3. Strong germline/ancestry component: Ewing incidence is markedly lower in African-ancestry populations and has multiple GWAS susceptibility loci - pointing to heritable, not environmental, drivers (PMC6085378; PMC12919121).
4. Arsenic's established human cancers are epithelial (bladder/lung/skin), not mesenchymal.

EWING_MECHANISM_MATRIX (summary; CSV rows added)

Bridge | Directness to Ewing | Level | Critical missing experiment

A HR suppression -> translocation-prone repair | indirect | d (strengthened) | Does env-relevant iAs raise DSBs / mis-repair at EWSR1 in MSCs?
 B genomic instability / LINE-1 | indirect | d | As dose vs structural-variant burden in mesenchymal progenitors
 C epigenetic/PRC2 shift | very indirect | d/e | Overlap of As epigenome with Ewing enhancer program
 D developmental window | indirect | d/e | Prenatal As timing vs fusion acquisition (model)
 Epi pesticide/farm signal | population, not arsenic-specific | c (pesticide) / g (As) | Re-analysis isolating arsenical exposure within "pesticide"

Verdict for the briefs

- Plausible, not proven. The mechanistic and epidemiological pieces are individually real; the arsenic->EWSR1 link is unstudied, and counter-evidence is substantive.
- The single highest-value experiment: test whether environmentally relevant inorganic arsenic increases EWSR1-region DSBs and/or biases their repair toward fusion in Ewing cells of origin (embryonic/pediatric mesenchymal progenitors) - see 16_proposed_experiments.

Sources (promote to MASTER_BIBLIOGRAPHY with full records + hashes)

- Arsenic & HR vs NHEJ - PubMed 37530740
- Oncogenomic disruptions / error-prone repair -> translocations - PMC5421966
- Arsenic & PRC2/H3K27me3 - PMC12223765
- MSC origin of Ewing - PMC2952797; Nat Commun 2025 s41467-025-64475-y
- Ewing farm/pesticide case-control - PMID 12020108; 1311140; 15551353
- CA perinatal PM & Ewing - PMC11887085
- Ewing GWAS loci - PMC6085378; PMC12919121

Cattle Dip Chemistry

CATTLE-DIP CHEMISTRY & SOIL FATE (Workstream 6 + fate link to 09)

Date: 2026-07-23 | single-reviewer. Not all arsenic compounds are interchangeable; the dip signature, toxicity, and soil behavior differ by species. Levels: a fact | b strong | c suggestive | d plausible | g unknown.

Chemical crosswalk (historical dip -> modern identity)

Historical name | Modern chemical | Formula | CAS | As valence | Notes

"White arsenic" / arsenic trioxide | Arsenic(III) trioxide | As_2O_3 | 1327-53-3 | +3 (trivalent) | the dip poison base; ~8 lb / 500 gal per USDA formula
Sodium arsenite | Sodium meta-arsenite | NaAsO_2 | 7784-46-5 | +3 | made from As_2O_3 + caustic soda; the active dip solution; highly soluble
Sodium arsenate | Disodium hydrogen arsenate | Na_2HAsO_4 | 7778-43-0 | +5 (pentavalent) | interconverts with arsenite in soil
Lead arsenate (contrast, orchard) | Lead hydrogen arsenate | PbHAsO_4 | 7784-40-9 | +5 | orchard pesticide - arsenic WITH lead; the other signature

Key toxicology (a/b): inorganic > organic; **trivalent arsenite ~10x more toxic than pentavalent arsenate; trivalent As binds critical sulfhydryl groups**, disrupting cellular metabolism.
Historical dip fluids reached ~200 mg As/L, acutely toxic to cattle that drank vat fluid.

The two signatures (decisive for testing - mirrors B5 board)

- Cattle-dip: arsenic trioxide -> arsenite/arsenate, low/absent lead -> *arsenic-with-little-or-no-lead* signature.
- Orchard: lead arsenate -> arsenic bound with lead signature.
- A lab distinguishes them by speciation + the As:Pb ratio. Local soil tests to date targeted the orchard/lead signature (see historical investigation); the dip signature has never been sought.

Co-contaminants historically present

Dip fluids/sites commonly co-occur with DDT (later additive) and, per formulation era, sulfur/lime; disposal and vat sludge concentrate arsenic. -> any local sampling must run **As speciation + lead + organochlorines** to separate sources (Workstream 13 alternatives).

Environmental fate at former dip sites (analog evidence, level b for the analog)

From studied former cattle-dip sites (chiefly Australia/South Africa; the same program the historical investigation compares against):

- > 1,000-1,600 former dip sites documented (NSW/QLD); a large legacy-contamination class.
- Surface-soil arsenic up to ~3,000 mg/kg (range ~34-2,941 mg/kg, 0-10 cm), **highest at the tank and drip-pad**; co-contaminated with DDT (up to ~60,000 mg/kg).
- Arsenic remains sequestered decades later, binding iron oxides and clays; **exceeds EPA residential screening levels**; groundwater exceedances reported; microbial effects persist 25 yr.
- Arsenic is an element - it does not degrade. Persistence is expected (a).

Relevance to the Ladera/Coto question (careful)

- Supports the source/pathway plausibility (d): if a vat operated locally, high, persistent, low-lead arsenic in the tank/drip-pad footprint is the expected signature - and it has **never been looked for** here.
- Does not establish local contamination (g): the analog sites are elsewhere; **only authorized local speciated soil sampling can confirm or exclude** a dip signature at these parcels.
- Confounder discipline: a high-As-with-lead result would point to orchard legacy, not the dip; a high-As-low-lead result is the dip signature. Speciation is the discriminator.

Sources

Chemistry/toxicity: Merck Vet Manual (inorganic arsenical toxicosis); inchem UKPID sodium arsenite;

As₂O₃?arsenite. Dip-site soil fate: PubMed 12798769; ScienceDirect S0160412006001395 (fractionation/bioaccessibility); Australia overview PMID 12635829. Formula (8 lb/500 gal) per the historical investigation's USDA Circular 174/183 records.

Biomarker Reconstruction Protocol

BIOMARKER EXPOSURE-RECONSTRUCTION PROTOCOL - E2 (DRAFT, gated)

Status: DRAFT for expert/IRB review. Not executed by this engine. Requires Ethics/IRB approval, individual consent, and chain of custody (GATE B). PHI stays in folder 04. Time-sensitive: teeth, nails, and archived newborn blood-spot cards are discarded/degrade on schedules. Levels: a fact | b strong | d plausible | g unknown.

Objective

Reconstruct individual historical inorganic-arsenic exposure, especially the **prenatal/early-life etiologic window** relevant to Ewing, in affected individuals vs matched comparisons - to test temporality and dose-gradient (the two Bradford-Hill elements currently at level g).

Rationale (validated methods)

- Deciduous ("baby") teeth: dentine grows in incremental layers; LA-ICP-MS reconstructs metal exposure at ~weekly resolution from 2nd trimester through early childhood (validated Pb/Mn/Ba/Sr; As feasible). PMID capture: S011.
- Toenails / fingernails: validated in-utero and long-term arsenic biomarker (S012; 19280039).
- Archived newborn dried blood-spot cards: state-banked at birth; enable methylation (255-CpG arsenic signature, S028) and metal analysis; used in childhood-cancer discovery (S013).
- Hair: recent months only; high external-contamination risk -> supportive, not primary.

Design

- Groups (blinded to lab): (i) affected individuals; (ii) unaffected community comparisons matched on age/sex/ancestry/residence-duration; (iii) external comparisons from a non-exposed area.
- Specimens per participant (as available/consented): shed deciduous teeth; toe/fingernail clippings; access to the archived newborn blood-spot card (state program, legal pathway).
- Timeline reconstruction: map dentine/nail As to calendar age; align to residence/school history (private, 04_) and the etiologic window (prenatal/early-life).

Analytical methods (indicative; lab-certified, blinded)

Specimen | Method | Readout

Tooth dentine | LA-ICP-MS micro-spatial + histology | weekly As (and Pb) trajectory, 2nd-trimester->early childhood
 Nails | trace-metal ICP-MS (acid digest) | integrated As over months; As species where feasible
 Blood-spot card | metal ICP-MS + EPIC methylation | birth As; 255-CpG arsenic exposure signature
 (all) | run lead alongside arsenic | separate dip vs orchard exposure context

Endpoints (pre-specified)

Primary: reconstructed As concentration/timeline in the etiologic window, affected vs comparison.

Secondary: presence of the arsenic methylation signature; As:Pb pattern; concordance across specimens.

QA/QC, ethics, custody

Field/reagent blanks, reference materials, duplicate analyses; blinded case/control labels until lock; documented chain of custody; IRB + individual consent; destructive testing minimized (teeth/ blood spots are finite - micro-sampling; retain remainder). No specimen handled by this engine.

Interpretation limits (state up front)

- A biomarker does not prove causation of a specific tumor; it establishes/?refutes?exposure and timing. Elevated + well-timed exposure strengthens; a null weakens the community-specific case.
- Contamination and metabolism (AS3MT genotype) confound - measure and adjust.

— - Small n -> report individual trajectories + exact CIs; avoid over-pooling.

Sequence & pairing

Run with E1 (soil): soil says was arsenic there; biomarkers say did this person carry it, when.

Together they are the strongest, cheapest early read on links (ii)-(v) before the hard mechanism work.

Falsification value (F2/F3)

Null reconstructed exposure across affected individuals would substantially weaken the hypothesis regardless of mechanism.

DRAFT - requires exposure-science + IRB review (GATE A/B) and consent before any specimen work.

11 TUMOR MOLECULAR SIGNATURES

Molecular Signature Protocol

TUMOR MOLECULAR SIGNATURES (Workstream 9) - feasibility + blinded protocol (design only)

Date: 2026-07-23 | single-reviewer | design for expert review; do not initiate testing (GATE B).

Levels: a fact | b strong | c suggestive | d plausible | g unknown.

Is there an arsenic signature, and can it be tested on Ewing?

- Candidate mutational signature (b): WGS of an arsenic-related lung tumor showed a **distinctive mutational spectrum** distinguishing arsenic-related from unrelated lung tumors of the same histology, with a high proportion of T>G transversions (oxidative-damage pattern). PMID 24128716.
 - Epigenetic exposure signature (b): a 255-CpG DNA-methylation signature predicts urinary arsenic, arsenical skin lesions, and mortality; ~1177 CpGs associate with urinary arsenic. PMID 42051231; PMC6143427.
 - The hard constraint (a): **Ewing sarcoma has one of the lowest mutation burdens of any cancer, ~0.15 mutations/Mb** (recurrent STAG2 ~21%, CDKN2A del ~14%, TP53 ~6%). PMID 25010205; 25223734.
- > Very few somatic mutations exist to carry a mutational signature. A T>G-signature approach is underpowered in Ewing; absence of the signature would be weakly informative, presence hard to reach significance.

Honest feasibility read

- Mutational-signature test: low power in Ewing (sparse mutations). Report as exploratory only.
- More feasible reads: (1) direct tumor arsenic (metallomics/LA-ICP-MS on FFPE) - measures the element, not a mutation count; (2) DNA-methylation profiling vs the arsenic CpG signature - epigenetic marks are abundant and FFPE-compatible on arrays; (3) **structural-variant / breakpoint microhomology** analysis at the EWSR1 junction (does the fusion breakpoint bear an error-prone-repair fingerprint?). None of these has been done for arsenic in Ewing -> all are gaps.

Blinded protocol (for GATE-B review; not initiated)

1. Samples: archived Ewing FFPE (community cases, with consent/legal authorization) + matched sporadic Ewing controls from outside the hypothesized area + non-Ewing sarcoma controls. Blinded to case/control status through analysis.
2. Assays: methylation array (EPIC), targeted/long-read sequencing of the EWSR1 breakpoint region, low-pass WGS for structural variants, and tumor metallomics (arsenic); run lead alongside.
3. Endpoints (pre-specified): (a) tumor arsenic vs controls; (b) presence of the 255-CpG arsenic methylation signature; (c) EWSR1 breakpoint microhomology/repair fingerprint; (exploratory) T>G mutational fraction.
4. Statistics: pre-registered; correct for multiplicity; power calc acknowledging low-mutation constraint; blinded code-break only after lock.
5. Interpretation limits (state up front): an arsenic exposure/methylation signature is **not proof of causation** of **this tumor**; signatures are non-specific and validated mainly outside sarcoma. A null does not exonerate; a hit is hypothesis-strengthening, not dispositive.

Chain of custody / ethics

FFPE blocks are finite and destroyable -> legal hold (Phase 3). Human-subjects + specimen approvals at GATE B. No specimen handling by this engine.

Sources

Arsenic lung mutational signature PMID 24128716; arsenic methylation signature PMID 42051231 / PMC6143427; Ewing genomic landscape / low burden PMID 25010205; 25223734.

11 TUMOR MOLECULAR SIGNATURES

Blinded Tumor Protocol

BLINDED TUMOR MOLECULAR & METALLOMIC PROTOCOL - E3 (DRAFT, gated)

Status: DRAFT for molecular-pathology + IRB review. Not initiated by this engine. FFPE blocks are finite/destroyable -> legal hold (Phase 3); human-subjects + specimen approvals at GATE B. Builds on MOLECULAR_SIGNATURE_PROTOCOL.md (09). Levels: a fact | d plausible | g unknown.

Objective

Test, blinded, whether Ewing tumors from the community differ from sporadic Ewing in (a) **tumor arsenic burden, (b) an arsenic exposure/methylation signature, and (c) EWSR1 breakpoint architecture** - while honestly bounding what each result can mean.

Hard design constraint (state up front)

Ewing has one of the lowest mutation burdens of any cancer (~0.15 mut/Mb) -> a **mutational-signature (e.g., arsenic T>G) approach is underpowered**; treat as exploratory only. Prioritize element (metallomics) and epigenetic/structural readouts, which do not depend on mutation count.

Samples (blinded; codes broken only after analysis lock)

- Arm 1 - community Ewing (FFPE, with authorization/consent).
- Arm 2 - sporadic Ewing controls from outside the hypothesized area (matched age/sex where feasible).
- Arm 3 - non-Ewing sarcoma controls (specificity).
- Adequate n per power analysis acknowledging the low-burden constraint; document block adequacy/tumor content; minimize destructive use (finite blocks).

Assays (FFPE-compatible; certified labs)

Assay | Readout | Note

Metallomics - LA-ICP-MS / ICP-MS on tumor + adjacent normal | tumor arsenic (and lead) | element, not mutation; primary
 EPIC methylation array | 255-CpG arsenic exposure signature (S028); tumor methylation class | epigenetic marks abundant in FFPE
 Targeted/long-read seq of EWSR1 breakpoint | breakpoint position + microhomology / repair fingerprint | does the fusion junction bear an error-prone-rep air signature?
 Low-pass WGS | structural-variant burden / genomic instability | context
 (exploratory) WES/WGS | T>G fraction, STAG2/TP53/CDKN2A status | low power - exploratory

Endpoints (pre-specified)

Primary: tumor arsenic (Arm 1 vs 2). Secondary: arsenic methylation signature presence; EWSR1 breakpoint microhomology vs sporadic. Exploratory: T>G mutational fraction; SV burden.

Statistics

Pre-registered; blinded; multiplicity correction; power calc stating the low-mutation limitation; report effect sizes + CIs, not just p-values.

Interpretation limits (governance)

- Exposure/methylation signatures are non-specific and not validated for causal attribution in sarcoma. A hit is hypothesis-strengthening, not proof; a null does not exonerate.
- Tumor arsenic could reflect post-diagnosis or agonal factors - interpret with exposure timeline (E2).
- Molecular-pathological-epidemiology framing: integrate with soil (E1) + biomarker (E2), not alone.

Ethics / custody

IRB; specimen consent; legal hold on blocks; chain of custody; results into private analytic layer (11_/13_ GIS), aggregated for any public output.

Sequence

Lower priority than E1/E2 (specimen-scarce, signatures non-specific). Run once blocks are secured and after E1/E2 give an exposure read to interpret against.

DRAFT - requires molecular-pathology + IRB review (GATE A/B) before any specimen work.

Epidemiological Analysis Plan

PREREGISTERED EPIDEMIOLOGICAL ANALYSIS PLAN (Workstream 10)

Date: 2026-07-23 | single-reviewer | hypothesis-neutral, no cluster asserted. This is a design to be registered before any observed counts are analyzed, per CDC/ATSDR guidance and to defuse the Texas-Sharpshooter (silent multiple-comparisons) problem. Levels: a fact | b strong | c suggestive | d plausible | e hypothesis | g unknown.

0. Governing framework

- CDC 2022 "Guidelines for Examining Unusual Patterns of Cancer and Environmental Concerns" - cancer cluster = "a greater-than-expected number of the same or etiologically related cancer cases in a group of people in a geographic area over a defined period." Uses a **10-criteria decision form; de-emphasizes statistical significance** as the gate; emphasizes community input and proactive registry review. (cdc.gov/cancer-environment/php/guidelines)
- Analysis must be authorized and use the California Cancer Registry (CCR), not tallies from media or social platforms (governance rule).

1. Case definition (pre-specified; do NOT pool biologically unrelated cancers)

- Primary: Ewing sarcoma - pathology-confirmed, EWSR1(FET)::ETS-fusion-positive, ICD-O-3 morphology 9260/3 (+ related PNET codes), any primary site (bone or extraskeletal), age 0-39 at dx.
- Secondary (separate analyses, never merged with Ewing): each other verified diagnosis handled in its own etiologically-defined stratum (osteosarcoma, leukemia, etc.). Rare-cancer heterogeneity is a known cluster-inflation trap; "any rare cancer" is not a valid case definition.

2. Geographic boundaries - PRE-SPECIFIED, with sensitivity set (not drawn around cases)

Primary: Ladera Ranch CDP. Sensitivity boundaries fixed in advance: (a) Ladera CDP; (b) Ladera + Covenant Hills; (c) 92694 ZIP; (d) 5-mile exposure ring (Zone B); (e) South-OC comparison set (Zone C: Talega, Aliso Viejo, Foothill Ranch - matched by development age/demographics, **never by case counts**). Boundaries are frozen before observed counts are examined.

3. Time boundaries

Primary dx window 2005-present (community build-out); extended 2000-present. Report by single years and rolling windows fixed in advance. Distinguish residence at diagnosis from residence during the etiologic window (for Ewing: prenatal/early-life - see §6).

4. Denominator & person-time (the usual failure point)

- Source: U.S. Census / ACS for the frozen boundaries; child/adolescent (0-19) and 0-39 strata; residency-duration and in/out-migration modeled (master-planned community -> high turnover).
- Person-years, not headcount, drive expected counts.

5. Expected incidence (SEER baseline, adjusted)

- Ewing ~2.93-3.0 per million/year, ages <20 (SEER). Peak 10-15 y; M:F ? 3:1; incidence ~9x higher in white than Black individuals. -> **age, sex, and ancestry standardization are mandatory* (an affluent, higher-white-proportion community raises the *expected baseline; failing to adjust would bias toward a false excess).

Illustrative expected-count math (FRAMEWORK ONLY - not an observed result)

For a hypothetical 0-19 population P over T years at rate $r = 2.93 \times 10^{-6}/\text{yr}$:

Expected $E = P \times r \times T$. Example $P \approx 10,000$, $T=20 \rightarrow E \approx 0.59$ cases; ancestry-adjusted upward

perhaps ~0.7-1.0. Interpretation caution: with $E < 1$, Poisson confidence intervals are very wide and

any single observed count is statistically unstable - a defensible analysis reports the **exact** Poisson/Byar CI and the SIR^{**}, never a point estimate alone, and never treats small numbers as proof. Observed counts must come from verified CCR case ascertainment, not this illustration.

6. Etiologic-window handling

Ewing likely initiates early (embryonic/pediatric mesenchymal progenitor; see 07_). Exposure-relevant residence is the prenatal/early-life window, which may differ from residence at diagnosis. Analyses that use only address-at-diagnosis will misclassify exposure.

7. Statistical methods

- Standardized Incidence Ratio (SIR) with exact Poisson (Byar) 95% CI; age/sex/ancestry standardized to CCR/SEER.
- Rare one-off cluster: report Bayesian posterior (weakly-informative prior) alongside frequentist, per the one-off-cluster literature (PMC2694210).
- Spatial/space-time: SaTScan (Kulldorff) Poisson/Bernoulli, circular and elliptic windows, multiple max-cluster scales (report all scans run), plus a Bayesian spatial model as cross-check.
- Small-area estimation for stability. Correct for multiple comparisons explicitly.

8. Bias controls

Ascertainment/verification bias (require pathology + CCR match; de-duplicate); ^{**}media-driven recruitment^{**} bias (register case-finding method); residential-mobility; boundary/multiplicity (pre-registration + report the search space); confounding by ancestry/SES.

9. Sensitivity analyses & negative controls

- Re-run across all §2 boundaries and §3 windows.
- Negative-control outcome: a common cancer with no arsenic hypothesis, analyzed identically - an excess there flags ascertainment artifact.
- Negative-control area: matched Zone-C community; expect null.

10. Analyses that would be INVALID or misleading (do not perform / do not cite)

- Cluster statistics from social-media / news case counts (governance red line).
- Post-hoc boundaries drawn to encircle known cases (Texas Sharpshooter).
- Pooling biologically unrelated rare cancers into one "cancer" count.
- Ignoring the denominator, person-time, or ancestry standardization.
- Reporting a point SIR for <5 cases without the exact CI and small-number caveat.
- Inferring causation from a spatial association (overlap != exposure != causation).

11. Preregistration & causal framing

Register this plan (boundaries, definitions, methods) before observed-count analysis. A statistically unusual pattern, if found, is a signal to investigate exposure, not evidence of arsenic causation; integrate with the source-pathway-receptor and Bradford-Hill work (14_causal_inference).

Sources

CDC 2022 guidelines (cdc.gov/cancer-environment); Ewing incidence SEER/ACS (StatPearls NBK559183; ACS key statistics); SaTScan/Kulldorff (satscan.org; PMC3965324); one-off cluster Bayesian PMC2694210; cluster causal-inference framework PMC8276584; challenges/limitations (CDC Challenges-and-Limitations PDF).

Preregistration Form

EPIDEMIOLOGY PREREGISTRATION FORM (DRAFT, to be time-stamped & locked BEFORE any observed counts)

Status: DRAFT. Purpose: register the design before analysts see case counts - the primary defense against the Texas-Sharpshooter / silent-multiple-comparisons problem. Complete, date/version-stamp, and lock (e.g., OSF or equivalent) prior to analysis. Derives from EPIDEMIOLOGICAL_ANALYSIS_PLAN. No social-media counts. Uses verified California Cancer Registry (CCR) data only.

1. Title / version / date-locked

[title] | v[x] | locked [YYYY-MM-DD HH:MM TZ] | registry [OSF/other ID]

2. Primary question (pre-specified, singular)

Is the age/sex/ancestry-standardized incidence of EWSR1::ETS-fusion-confirmed Ewing sarcoma in [frozen geography] during [frozen period] greater than expected from CCR/SEER reference rates?

3. Case definition (frozen)

- Ewing: pathology-confirmed, EWSR1(FET)::ETS fusion-positive, ICD-O-3 9260/3 (+ specified PNET codes), ages [0-39], bone or extraskelatal. Ascertained via CCR; de-duplicated; pathology verified. Secondary cancers analyzed separately (list): [...]. No pooling of unrelated cancers.

4. Geography (frozen; sensitivity set fixed now)

Primary [Ladera Ranch CDP]; sensitivity [+Covenant Hills], [92694 ZIP], [5-mi ring], control [Zone-C matched community]. Boundaries frozen before counts examined.

5. Time window (frozen)

Primary [2005-present]; extended [2000-present]. Etiologic-window residence (prenatal/early-life) captured where available; distinguished from residence-at-diagnosis.

6. Denominator / person-time (pre-specified sources)

Census/ACS for frozen boundaries; strata 0-19 and 0-39; residency-duration + migration modeled.

7. Expected-rate reference & standardization (pre-specified)

SEER/CCR Ewing rates; standardize by age, sex, and ancestry (Ewing ~9x higher in white). State the reference tables/years now.

8. Primary statistic + secondary

SIR with exact Poisson/Byar 95% CI (primary); Bayesian posterior with prior [specify] (secondary); SaTScan Poisson/Bernoulli, circular+elliptic, max cluster sizes [list], all scans reported; small-area estimation. Multiplicity handling [method].

9. Case-finding method (register to expose recruitment bias)

[CCR query only / active outreach / ...] - pre-declare; flag any media-driven ascertainment.

10. Sensitivity analyses (pre-listed)

All §4 boundaries x §6/§7 assumptions; exclude/include borderline diagnoses; migration assumptions.

11. Negative controls (pre-specified)

Outcome control: [common non-arsenic cancer] analyzed identically (expect null).

Area control: [Zone-C] (expect null).

12. Decision rules (pre-specified; significance de-emphasized per CDC 2022)

State in advance what each result triggers (e.g., SIR + CI ranges -> "further exposure assessment" vs "no further action"), per the CDC 10-criteria framework - not a bright-line p-value.

13. Analyses declared INVALID (will not be run/reported)

Social-media/news counts; post-hoc boundary drawing; pooling unrelated cancers; denominator/ancestry neglect; point SIR for <5 cases without exact CI + small-number caveat; causal inference from spatial overlap alone.

14. Data governance

CCR authorization [status]; PHI in folder 04; public outputs masked/aggregated; analysts blinded to hypothesis where feasible; code + seed archived for reproduction.

15. Authors / reviewers / roles

[epidemiologist, biostatistician, registry liaison, physician lead]. Dual-review before lock.

DRAFT - complete, lock, and time-stamp before any count is analyzed. GATE A (design review) -> lock -> authorized CCR analysis.

Gis Schema

GIS SCHEMA (Workstream 11) - spatial/temporal integration

Privacy first: exact case residences are PHI (folder 04, gitignored). Public/general layers use aggregation, jittering, or areal masking (e.g., counts per $\geq$ block-group, or distance-to-source bands) - never point locations. Analytic (private) layers may hold exact points under access control + chain of custody only.

Coordinate/analysis standard

CRS EPSG:2230 (CA State Plane VI, ftUS) for local work; store WGS84 too. Document positional accuracy and source per feature (provenance rule).

Layers (each: geometry, source_id, grade, date/coverage, accuracy, privacy_class)

A. Health (PRIVATE - 04 only)

- Case residence(s) by exposure window (prenatal/early-life/at-dx); school/daycare attendance; diagnosis year; verified dx + EWSR1 status. Public derivative = masked aggregate only.

B. Historical exposure geography

- Documented 1908 Joplin dip (Trabuco/Bell Canyon); probable dip grounds; corrals/chutes; livestock routes; historic ranch structures; drainages; historic roads (from the historical investigation).

C. Development / earth-movement (exposure redistribution)

- Grading & cut-and-fill plans; development phases (1999-2006); imported/exported soil; utility corridors; foundation excavation; dust-generating construction footprints/timeline.

D. Environmental measurements

- Soil-sampling points + results (total & bioaccessible As, lead, speciation, 0-6 in horizon); background/natural-arsenic geology; groundwater & surface water; wind climatology (dust pathway).

E. Receptor/land-use

- Schools, daycares, parks, playgrounds, common areas, greenbelts, recycled-irrigation zones.

F. Confounders/co-sources

- Former orchards (lead-arsenate signature), agricultural chemical use, oil & gas, other point sources.

Core spatial questions (analysis, not proof)

1. Distance from masked case aggregates to documented/probable dip grounds vs matched controls.
2. Overlap of child-contact receptors (D/E) with historical exposure geography (B) and earth-movement (C).
3. Where soil sampling (D) should target - feeds 14_proposed_experiments & the epi denominators.

Outputs

- Public: masked/aggregated maps only, each carrying the LEHRP-style neutrality caveat ("geographic overlap != exposure != causation").
- Analytic (private): exact-location layers, access-controlled.

Sources

Historical-investigation geospatial layers (repo research/..., data/geospatial/); Census/ACS boundaries; CA DTSC/EnviroStor; USGS geology; grading/PEIR records (per archive list).

14 CAUSAL INFERENCE

Causal Weight Of Evidence

CAUSAL WEIGHT OF EVIDENCE (Workstream 12)

Date: 2026-07-23 | single-reviewer (flag dual review) | hypothesis-neutral. Applies multiple frameworks (Bradford Hill, source-pathway-receptor, mode-of-action) to the question **"did historical inorganic arsenic cause Ewing sarcoma in this community?"** - not as a checklist that yields a verdict, but to show which links have evidence and which do not. Levels: a fact | b strong | c suggestive | d plausible | e hypothesis | f inference | g unknown.

Headline

On the specific question (arsenic -> Ewing), the weight of evidence is currently ****weak / unestablished**: most Bradford-Hill considerations are unmet or unstudied**, plausibility is moderate but mechanistically uncertain, and there is real incoherence (tissue tropism + dominant arsenic mode of action). The environmental predicate (arsenic can persist at high levels at former dip grounds) is comparatively well-supported by analog sites.

Bradford-Hill considerations - arsenic -> Ewing

Consideration | Evidence for | Evidence against | gap | Level

Strength of association | none for As-Ewing; farm/pesticide->Ewing RR up to ~6 (not As-specific) | no direct As-Ewing estimate exists | g (As); c (pesticide)
 Consistency | pesticide/farm signal replicated across countries | no As-Ewing studies to be consistent | g
 Specificity | - | arsenic is non-specific (many cancers); Ewing has one defining lesion of unknown cause | weak both ways
 Temporality | biomarkers could establish exposure-before-disease | not yet measured in any case | g
 Biological gradient | dose-response established for As's known (epithelial) cancers | none for Ewing | g for Ewing
 Plausibility | HR-suppression/error-prone-repair -> translocation link (07_) | dominant As MOA is cytotoxicity + regenerative proliferation / oxidative stress, i.e. often classed non-genotoxic - not translocation-formation | d, tempered
 Coherence | arsenic persists in dip-site soil (below) -> exposure coherent | As tissue tropism is epithelial, not mesenchymal bone | mixed
 Experiment | - | the decisive EWSR1-break/fusion experiment has never been run | g (key gap)
 Analogy | hepatic angiosarcoma = As causes a human sarcoma | APL: As2O3 treats a fusion cancer (counter-analogy) | c / counter

Source-Pathway-Receptor

Link | Status | Best support | Gap

Source - arsenic present at dip ground | plausible; 1908 Joplin dip documented nearby | analog dip sites retain 34-2,941 mg/kg As decades on | not measured at these parcels
 Pathway - retention -> redistribution -> child contact | plausible | As sequesters on Fe-oxides/clay, highest at tank/drip pad; grading could remobilize | no local soil/dust data
 Receptor - children exposed at dose relevant to Ewing | unknown | biomarkers recoverable | no dose data; no As-Ewing dose model
 Outcome - that exposure caused these cancers | unestablished | - | specific causation unproven

Mode-of-action check (why plausibility is tempered)

Arsenic's accepted MOA for its established cancers centers on ****oxidative stress + cytotoxicity with regenerative proliferation*** (often described as ***non-genotoxic carcinogenesis***), plus epigenetic and repair effects. That MOA does not map cleanly onto "generate a specific EWSR1 translocation in a mesenchymal progenitor." The genotoxic/HR-suppression thread (Bridge A) is a minority mechanistic route. ****Honest read: the mechanism most responsible for arsenic's proven cancers is not the mechanism the Ewing hypothesis requires**** - a genuine weakness to state plainly.

Triangulation across independent streams

- Mechanistic (07_): plausible-not-proven.
- Epidemiologic (12_): Ewing-pesticide signal real but not arsenic-resolved; local cluster unverified.
- Exposure/environmental (below, 09_): dip-site arsenic persistence well-analogized; local unmeasured.

— - Molecular (09_): no arsenic signature tested against Ewing.

-> The streams converge on plausibility and on a shared set of decisive, unrun tests, not on demonstrated causation.

What would move the needle (feeds 16_)

Establish temporality + gradient via biomarker exposure reconstruction; experiment via EWSR1-break/fusion assays under environmentally relevant arsenic; coherence via local soil speciation (dip signature vs orchard); molecular via blinded tumor signature testing.

General vs specific causation

- General: arsenic -> Ewing is not established (unstudied); arsenic -> a human sarcoma is established (angiosarcoma).
- Specific: unestablished; requires local arsenic + reconstructed individual exposure in the etiologic window + an accepted mechanism/biomarker link.

Sources

Arsenic MOA / non-genotoxic: ScienceDirect S1001074216301255; S0041008X06003632; review PMID 30223072.

Dip-site soil arsenic persistence: PubMed 12798769; S0160412006001395; Australia overview PMID 12635829.

(Mechanism + epi + biomarker sources per 07_/12_/10_.)

14 CAUSAL INFERENCE

General Vs Specific Causation

GENERAL vs SPECIFIC CAUSATION - framework & current status (DRAFT, gated)

Status: DRAFT scientific framework for the physician + attorney teams. Not a legal conclusion.

This engine draws no legal conclusions; it maps where evidence does and does not exist along each causal element. Levels: a fact | b strong | c suggestive | d plausible | e hypothesis | g unknown.

The two questions (kept strictly separate)

- General causation: Can inorganic arsenic, at environmentally relevant exposure, cause Ewing sarcoma (or another reported cancer) in humans, in principle?

- Specific causation: Did arsenic exposure cause this individual's cancer - i.e., were they exposed, at a sufficient dose, during the etiologic window, before diagnosis, with alternatives reasonably excluded?

Specific causation cannot be reached unless general causation is first supported. Neither is an argument this engine makes; both are evidentiary maps.

General causation - current status

Element | Status | Level | Basis

Arsenic is a human carcinogen | yes | a | IARC Group 1

...for Ewing / bone sarcoma | not established | a (for the absence) | no sufficient human evidence; established sites are epithelial

Direct As->Ewing etiologic study | none exists; only As-as-therapy | a (absence + counter-signal) | native-db: 0 hits; APL therapy is opposite direction

Plausible mode of action to a fusion sarcoma | partial, non-specific | d | HR suppression/instability -> translocations in general; not shown at EWSR1

Dose-response for Ewing | unknown | g | never studied

General-causation verdict (today): NOT SUPPORTED for Ewing. Biological plausibility exists at a low level; the affirmative human evidence does not. This can change only with the mechanistic and epidemiologic work in Phases 2-4.

Specific causation - elements & what would be needed (per individual, private/04)

Element | Evidence needed | Currently

Exposure occurred | soil at residence/school + biomarker (E1/E2) | unmeasured

Sufficient dose | reconstructed dose vs any threshold | unknown

Correct timing (etiologic window) | tooth/nail/blood-spot timeline vs residence history | unmeasured

Temporality (exposure before disease) | exposure reconstruction pre-diagnosis | unmeasured

Alternatives excluded | Workstream-13 confounders (orchard lead-arsenate, DDT, ancestry, chance) | not done

Individual biological link | blinded tumor As/signature (E3) - hypothesis-strengthening only | unmeasured

Specific-causation verdict (today): CANNOT BE ASSESSED - the predicate measurements do not exist, and general causation is not yet supported.

What this means for the teams (neutral)

- The honest current position is "unmeasured and unestablished," not "refuted." The value of Phases 2-4 is that they can move each cell above from g/d toward a/b in either direction.

- Any public or legal statement must respect these levels; do not state or imply specific causation.

- Confounding is central: former-orchard lead-arsenate and DDT co-occur with the dip-era arsenic and with the documented pediatric-cancer environmental signal (pesticides) - separating them (speciation, As:Pb, molecular) is prerequisite to any specific-causation reasoning.

Falsification (either direction advances the science)

- General causation weakened if: mechanistic tests show no EWSR1-region effect (F-series), and/or exposure reconstruction is null across affected individuals.

- Specific causation weakened for an individual if: no exposure/dose in the etiologic window, or a competing cause is established.

DRAFT - for expert + legal review at GATE A/B. No legal conclusion is expressed or implied.

15 ALTERNATIVE HYPOTHESES

Alternative Hypotheses

ALTERNATIVE & CO-EXPOSURE HYPOTHESES (Workstream 13)

Date: 2026-07-23 | single-reviewer. Arsenic is the primary hypothesis; these are evaluated for scientific comparison and confounding control, not to dilute it. Each verified case would be assessed against all. Levels: a fact | c suggestive | d plausible | g unknown.

Alternative / co-exposure | Relationship to reported cancers | Ewing-relevant evidence | Discriminator from arsenic

Lead-arsenate (orchard) | arsenic with lead | orchard/pesticide->Ewing signal (12_) | As:Pb ratio + speciation (high lead = orchard)

Organochlorines (DDT/chlordane/DDE) | co-contaminant at dip AND orchard sites | pesticide->Ewing case-control signal may reflect these | organochlorine panel in soil/serum

Organophosphates / other pesticides | agricultural signal | "pesticide" studies rarely speciate | targeted residue analysis

Other metals (lead, chromium, beryllium) | co-occur (e.g., SOLAG site) | lead neurodev, not Ewing | metals panel

Petroleum / oil & gas | pediatric-cancer literature; a CA Ewing OGD study exists | proximity study (mixed/null) | distance-to-well; PAHs

PFAS, nitrates, radon, air pollution (PM) | general | a CA perinatal-PM->Ewing study exists (PMC11887085) | media-specific measurement

Parental occupational exposure | replicated Ewing signal (paternal ag/solvents; maternal pesticide) | strongest environmental Ewing evidence (12_) | occupational history, not soil

Germline predisposition / ancestry | Ewing has GWAS loci; 9x white vs Black | competing driver | germline testing; ancestry adjustment

Prior therapeutic radiation/chemo | known sarcoma cause | secondary sarcoma | treatment history

Chance aggregation | small-number rare-cancer clustering | Texas-Sharps shooter risk | preregistered epi (10_)

Ascertainment / media-recruitment / duplicate reports | inflates apparent counts | - | verified CCR ascertainment

How this constrains the arsenic case

- The single most important confounder is the orchard lead-arsenate signature - many local/near sites tested positive for arsenic with lead (the orchard kind). **Only low-lead speciation distinguishes a dip source from an orchard source.**
- The replicated environmental Ewing signal is "pesticide/farm/parental-occupation," not arsenic - arsenic has never been isolated within it. A defensible analysis must **model organochlorines and parental occupation** before attributing to arsenic.
- Ancestry/germline must be adjusted (Workstream 10) or an apparent excess in an affluent, higher-white community is partly expected baseline.

Rule

Do not let alternatives dilute the arsenic investigation - but **no arsenic attribution is defensible until these are measured and excluded/adjusted** at the individual and area level.

16 PROPOSED EXPERIMENTS

Proposed Experiments

PROPOSED CRITICAL EXPERIMENTS (Workstream 14) - design only; execute only after GATE B

Date: 2026-07-23 | single-reviewer. Ranked by evidentiary value / feasibility. **No experiment is initiated by this engine.** Environmentally relevant doses only - do not substitute unrealistic high-dose work. Levels of current evidence they address in brackets.

Tier 1 - resolve the local exposure predicate (fastest, cheapest, decisive-ish)

E1. Authorized speciated soil sampling at priority grounds [source/pathway: d->?]

- Hypothesis: a low-lead arsenic-trioxide ("dip") signature exists above background at the documented/probable dip ground (Bell Canyon/Starr Ranch vicinity) and/or child-contact areas.
- Design: 0-6 in child-contact horizon + depth profiles at tank/drip-pad analogs; **total + bioaccessible As, lead, speciation, organochlorines**; background controls outside cattle-use area.
- Endpoints: As concentration; As:Pb ratio + speciation (dip vs orchard); bioaccessibility.
- Value: high - converts a paper hazard to measured presence/absence. Ethics: landowner + agency permission (Audubon for Starr Ranch); chain of custody. Limit: mass grading may have diluted signal.

Tier 2 - reconstruct individual historical exposure (temporality + gradient)

E2. Biomarker exposure reconstruction [temporality: g->?]

- Deciduous-tooth dentine LA-ICP-MS (weekly-resolution prenatal->early-childhood As), infant/adult toenails, archived newborn dried blood-spot metals/methylation, in affected vs matched comparison.
- Endpoints: reconstructed As timeline vs etiologic window; the 255-CpG methylation signature.
- Value: high; the only route to exposure-before-disease. Ethics: IRB + specimen consent; PHI.

Tier 2 - tumor molecular (feasibility-limited; see 11_)

E3. Blinded Ewing FFPE metallomics + methylation + EWSR1-breakpoint analysis

- Endpoints: tumor arsenic; arsenic methylation signature; breakpoint microhomology/repair fingerprint; (exploratory) T>G fraction - with the low-mutation-burden caveat.
- Value: medium (non-specific signatures; sparse mutations). Ethics: finite blocks; legal hold.

Tier 3 - the mechanistic crux (arsenic -> EWSR1)

E4. EWSR1 double-strand-break induction - env-relevant iAs in human embryonic/pediatric mesenchymal progenitors; measure DSBs (γH2AX, breakpoint PCR) at/around EWSR1. [plausibility d->test]

E5. Repair-fidelity assay - does iAs bias repair of an induced EWSR1-region break toward translocation (HR-suppression readout, given As lowers HR not NHEJ)?

E6. Fusion-formation / selection assay - does iAs increase de-novo EWSR1::FLI1 formation, or select/expand cells already carrying it? (recall EWS-FLI1 alone needs cooperating events)

- Value: highest for general causation mechanism; Feasibility: hard; specialized labs.
- Controls: vehicle; positive genotoxin; APL counter-model (arsenic kills fusion cells) to bound effect direction.

Tier 3 - population

E7. Preregistered CDC/ATSDR epidemiology (Workstream 10) with verified CCR counts, ancestry standardization, SaTScan + Bayesian; E8. gene-environment study of arsenic-metabolism variants (AS3MT) in cases; E9. comparison population near a documented dip-contaminated site.

Prioritization logic

Do E1 + E2 first (cheapest, resolve exposure predicate + temporality). E4-E6 answer the mechanism but are slow. E7 needs authorized registry data. Every experiment: pre-specified endpoints, blinding where possible, replication, confounder control (13_), ethics + chain of custody.

Falsification value (ties to TOP_15)

Null E1 (no dip signature) + null E2 (no reconstructed exposure) would substantially weaken the community-specific hypothesis without needing the hard mechanism work.

16 PROPOSED EXPERIMENTS

Soil Sampling And Analysis Plan

SOIL SAMPLING & ANALYSIS PLAN - E1 (DRAFT, pending GATE A review + GATE B authorization)

Status: DRAFT for expert/agency review. Not executed by this engine. Real sampling requires landowner/agency permission + chain of custody (GATE B). Time-sensitive: signal is lost once ground is graded/remediated. Levels: a fact | d plausible | g unknown.

Objective

Determine whether a low-lead arsenic-trioxide ("cattle-dip") signature exists above local background at documented/probable dip grounds and at child-contact areas - and **distinguish it from the orchard lead-arsenate signature*. Resolve exposure link (ii) from *plausible toward measured.

Hypotheses tested (from TOP_20)

H1 dip signature above background at the documented dip ground; H2 dip signature in child-contact areas; plus confounder test (orchard lead-arsenate; organochlorines).

Priority sampling areas (pre-specified)

1. Bell Canyon / Starr Ranch vicinity - the documented 1908 Joplin dip ground; apparently undeveloped (Audubon sanctuary) -> cleanest test. Highest priority.
2. Ladera Ranch preserved Trabuco greenbelt + neighborhood parks/common areas (child-contact).
3. Background/control locations outside any cattle-use/orchard area, matched soil type.
4. Optional: a matched Zone-C community (negative-control area).

Design

- Depth: 0-6 in child-contact horizon (primary) + depth profiles (0-6, 6-12, 12-24, 24-36 in) at suspected tank/drip-pad analogs to capture a buried plume.
- Spatial: systematic grid + targeted points at geomorphic drain/pen features; **model-based geostatistics** for spatial variability (per dip-site literature); georeference each point (private if near a residence).
- QA/QC: field duplicates, equipment blanks, trip blanks, matrix spikes; documented chain of custody; certified lab.

Analyte suite (the discriminator is the whole point)

Analyte | Method (indicative) | Why

Total arsenic | EPA 6010/6020 (ICP-MS) | concentration vs background (~<=12 mg/kg SoCal upper bound)
 Total lead | EPA 6010/6020 | As:Pb ratio - dip (low-Pb) vs orchard (As+Pb)
 Arsenic speciation (As(III)/As(V)) | IC-ICP-MS / XANES | source + toxicity form
 Bioaccessible arsenic | in-vitro bioaccessibility (e.g., SBRC/UBM) | hazard->dose realism (not total alone)
 Organochlorines (DDT/DDE/DDD, chlordane) | EPA 8081 | co-contaminant / alternative source (13_)
 Iron/Mn oxides, pH, OM, texture | standard soil | As binding/mobility context

Interpretation rules (pre-specified)

- Dip signature: arsenic >> background with low/absent lead (low As:Pb) -> supports a dip source.
- Orchard signature: arsenic elevated with co-elevated lead (lead-arsenate) -> points to orchard, not the dip (confounder wins).
- Organochlorine co-elevation -> agricultural source mixture; weigh in 13_.
- Null (<= background): does not fully clear the site (grading may have diluted/removed surface residue) - interpret with the site-disturbance history; still substantially weakens H1/H2.
- Report totals and bioaccessible fractions; never infer dose from total alone.

Screening references (context, not verdicts)

SoCal ambient arsenic mean ~1.5 mg/kg, upper bound ~12 mg/kg (school-PEA dataset); WA former-orchard cleanup >20 ppm; former dip sites elsewhere reach 34-2,941 mg/kg near tank/drip-pad. Use local background from the control points, not a fixed number.

Ethics / custody / sequence

Authorized + permitted only (Audubon for Starr Ranch); chain of custody; results into the GIS (11_, exact points private). Do this before any grading/remediation at priority grounds.

Falsification value

Null dip-signature here (F1) removes a core pillar of the community-specific hypothesis at low cost - pair with biomarker reconstruction (E2) for the strongest early read.

DRAFT - requires environmental-toxicology + geochemistry review (GATE A) and landowner/agency + legal authorization (GATE B) before any field work.

Immediate Evidence Preservation Checklist

IMMEDIATE EVIDENCE-PRESERVATION CHECKLIST (time-critical)

Why this is first: several evidence streams that could reconstruct exposure *years before diagnosis* or fix the environmental record are destroyable or degrade with time. Preserving them does not assume causation - it protects the ability to test the question either way. Physician/legal teams should secure chain of custody; this project does not collect specimens or contact families.

Level key: a fact | b strong | c suggestive | d plausible | e hypothesis | g unknown.

A. Biological specimens (can reconstruct historical/in-utero arsenic) - HIGH urgency

Specimen | What it reconstructs | Evidence it works | Risk of loss

Deciduous ("baby") teeth | Weekly-resolution metal exposure from ~2nd trimester through early childhood via dentine growth rings (LA-ICP-MS) | validate d for Pb/Mn/Ba/Sr; As feasible - Nature JESEE s41370-021-00400-x; ScienceDirect S0160412021004748 | teeth discarded; families may still have them - act now

Infant/adult toenails, hair | Long-term + in-utero arsenic (nails a validated As biomarker) | Nature jes201438; PubMed 19280039 | routinely discarded

Archived newborn dried blood-spot cards | State-banked at birth; enable unbiased early-life biomarker + methylation studies for childhood cancer | Clin Chem 2026 (72:1:82); cord-blood As methylation PMC3506565 | state retention/consent windows; destroyed on schedules

Archived FFPE tumor blocks / surgical specimens | Tumor arsenic + molecular signatures (Workstream 9) | pathology archives standard | hospital retention limits; blocks discarded/exhausted

Placental tissue, deciduous teeth of siblings (controls) | comparison/exposure timing | - | discarded

B. Environmental samples - HIGH urgency (irreversible if disturbed)

- **Soil at documented/probable dip ground (Bell Canyon / Starr Ranch vicinity) and school/park play areas - BEFORE any grading, remediation, or development disturbs it.** Test total + bioaccessible arsenic, with lead, plus speciation (to separate dip-signature arsenic trioxide from orchard lead-arsenate). Once soil is moved/capped, the depth profile and signature are lost.
- Any predevelopment / Phase-I era soil archives or boring logs for Ladera and Coto tracts.
- Chain-of-custody sampling only; authorized + landowner-permitted (Audubon for Starr Ranch).

C. Documentary/records - MEDIUM urgency (see historical investigation's archive list)

- Rancho Mission Viejo / O'Neill ledgers, foreman logs, dipping receipts (Sherman Library; private RMV archive); County Veterinarian orders + Supervisors minutes (OC Archives); State Vet files (CA State Archives). These could name a vat location. Private archives can deaccession.

D. Registry/medical (private, folder 04) - MEDIUM urgency

- Confirm pathology + EWSR1 fusion status/partner for each reported Ewing case; residence/school exposure windows; prenatal/parental residence - captured privately, aggregated for any public output. Verify diagnoses against California Cancer Registry (avoid social-media case counts).

First actions for the clinical/legal team

1. Ask affected families (via their own clinicians/counsel, not this project) to retain baby teeth, nails, and any keepsakes before disposal.
2. Petition to preserve archived newborn blood-spot cards and FFPE blocks under legal hold.
3. Secure authorized soil sampling with chain of custody at priority grounds before any ground disturbance; analyte set = total+bioaccessible As, lead, speciation.
4. Place document holds on ranch/county/state archives that could name a vat.

20 RESEARCH GAPS

Top 25 Critical Research Gaps

TOP 25 CRITICAL RESEARCH GAPS

- 1 No study of arsenic AND Ewing sarcoma (any design).
- 2 Whether env-relevant iAs raises EWSR1-region DSBs.
- 3 Whether iAs biases EWSR1 break repair toward fusion.
- 4 Whether iAs selects/expands EWSR1::FLI1+ progenitors.
- 5 Local soil arsenic - concentration + speciation (dip vs orchard) - never measured.
- 6 Local soil organochlorines/lead (confounders) - unmeasured.
- 7 Individual historical arsenic exposure in cases (teeth/nails/blood spots) - never reconstructed.
- 8 Tumor arsenic (metallomics) in local Ewing - never measured.
- 9 Arsenic methylation signature tested in Ewing - never.
- 10 EWSR1 breakpoint repair-fingerprint analysis - never.
- 11 Verified (CCR) case count + denominator for the community - not established here.
- 12 Ancestry/SES-standardized expected incidence for this community - not computed with real data.
- 13 Etiologic-window residence histories - not assembled (privately).
- 14 Dose-response for arsenic and any sarcoma - unknown.
- 15 Whether grading redistributed any dip-era arsenic - unmodeled.
- 16 Bioavailability of any local arsenic (RBA) - unmeasured.
- 17 Parental occupational-exposure histories for cases - not collected.
- 18 Gene-environment (AS3MT metabolism variants) in cases - unstudied.
- 19 Groundwater/recycled-irrigation arsenic pathway locally - unassessed.
- 20 Comparison population near a documented dip-contaminated site - not studied.
- 21 Whether "pesticide" Ewing studies captured arsenical exposure - needs re-analysis.
- 22 Prenatal timing of EWSR1 fusion acquisition in humans - unresolved.
- 23 Native-database (PubMed/EuropePMC) exhaustive counts for all Workstream-3 strings - pending.
- 24 Dual-reviewer verification of consequential findings - pending (GATE A).
- 25 Any historical predevelopment environmental sample from these parcels - located? unknown.

Top 20 Testable Hypotheses

TOP 20 TESTABLE HYPOTHESES (each -> an experiment in 16_)

- H1 A low-lead arsenic-trioxide soil signature exists above background at the documented dip ground.
- H2 Child-contact areas (parks/schools/yards) carry the dip signature.
- H3 Affected individuals show elevated reconstructed prenatal/early-childhood arsenic (teeth/nails).
- H4 Archived newborn blood spots of cases carry an arsenic methylation signature.
- H5 Local Ewing tumors contain elevated arsenic (metallomics) vs sporadic controls.
- H6 EWSR1 breakpoints in local cases show error-prone-repair microhomology signatures.
- H7 Env-relevant iAs increases DSBs at/near EWSR1 in mesenchymal progenitors.
- H8 iAs suppresses HR repair of an induced EWSR1 break (consistent with S003).
- H9 iAs increases de-novo EWSR1::FLI1 formation frequency in vitro.
- H10 iAs selects survival/expansion of EWSR1::FLI1+ cells.
- H11 Community Ewing incidence exceeds ancestry-standardized expectation (verified CCR).
- H12 Cases cluster spatially near dip/ranch geography more than controls (masked).
- H13 Cases have longer residence in the etiologic window than controls.
- H14 AS3MT (arsenic-metabolism) risk genotypes are over-represented in cases.
- H15 Grading redistributed dip-era arsenic into residential surface soil.
- H16 Local arsenic is bioaccessible (RBA meaningfully > 0).
- H17 Organochlorine co-exposure independently associates with cases (confounder test).
- H18 Parental agricultural/pesticide occupation is over-represented (replicates prior signal).
- H19 A matched Zone-C community shows null soil signature and expected incidence.
- H20 Populations near documented dip-contaminated sites show excess sarcoma (external test).

Top 15 Potential Falsification Findings

TOP 15 FINDINGS THAT WOULD FALSIFY / MATERIALLY WEAKEN THE HYPOTHESIS

- F1 No low-lead arsenic signature above background in local soil (H1/H2 null).
 - F2 No elevated reconstructed arsenic in affected individuals (H3 null).
 - F3 No arsenic methylation signature in newborn blood spots of cases (H4 null).
 - F4 Local Ewing tumors show no elevated arsenic and typical sporadic genomics (H5 null).
 - F5 EWSR1 breakpoints show no repair-fingerprint distinct from sporadic Ewing (H6 null).
 - F6 Env-relevant iAs does not increase EWSR1 DSBs/fusion/selection (H7-H10 null).
 - F7 Ancestry-standardized incidence is within expected range on verified counts (H11 null).
 - F8 No spatial association of cases with dip/ranch geography (H12 null).
 - F9 Cases predominantly have known germline predisposition or prior radiotherapy.
 - F10 Any apparent excess is explained by ascertainment/media-recruitment/duplication.
 - F11 High local arsenic is the lead-arsenate (orchard) signature, not the dip (confounder wins).
 - F12 Organochlorine/parental-occupation exposure fully accounts for any case pattern.
 - F13 Local arsenic present but non-bioaccessible (no plausible dose).
 - F14 Grading demonstrably buried/diluted any residue below contact-relevant levels.
 - F15 Comparison populations near documented dip contamination show NO sarcoma excess.
- Any of F1-F8 would remove a core pillar; F1+F2 together would substantially weaken the community-specific case even before mechanism work.

Contradictory Evidence

CONTRADICTORY / WEAKENING EVIDENCE (weighted equally with supportive)

Consolidated so the case against the arsenic->Ewing hypothesis is as visible as the case for it.

1. No direct study links arsenic to Ewing (05_). Absence of research, but also absence of support.
2. **Arsenic's dominant MOA is cytotoxicity/regenerative proliferation / oxidative stress (often non-genotoxic)** - not translocation-formation (14_; S021,S022). The mechanism the hypothesis needs is a minority route.
3. Tissue-tropism mismatch: arsenic's established cancers are epithelial (bladder/lung/skin); Ewing is mesenchymal bone (06_).
4. Counter-analogy: arsenic trioxide cures APL, a fusion-driven leukemia - arsenic can destroy translocation-cancer cells, not only cause them (05_,07_).
5. EWS-FLI1 alone does not transform normal human cells; cooperating events are required, and the fusion itself impairs DNA repair - the rate-limiting step may not be break-formation (07_,S006).
6. Strong germline/ancestry component: GWAS loci; ~9x higher in white than Black individuals - points to heritable drivers and demands ancestry standardization (10_,S014,S017).
7. Ewing's very low mutation burden (0.15/Mb) limits power to detect an arsenic mutational signature in the tumor (09_,S25010205).
8. The replicated environmental Ewing signal is "pesticide/farm/parental-occupation," not arsenic; arsenic was never isolated, and organochlorines co-occur (12_,13_).
9. Local exposure unmeasured: dip-site arsenic persistence is shown elsewhere; no local soil, dust, or biomarker data exist (08_,14_).
10. Small-number/Texas-Sharpsooter risk in any community cluster claim (10_).

Net: these do not disprove the hypothesis, but they make causation currently unsupported and define why measurement (not assertion) is the only defensible path.

11. Directional counter-signal (native-db, reproducible): the entire published arsenic-Ewing literature is arsenic trioxide as a treatment (it kills Ewing cells, like it treats APL) - arsenic AND EWSR1 = 0, arsenic AND EWS-FLI1 = 0 in PubMed. Zero etiologic studies exist.

Next 30 Priority Actions

NEXT 30 PRIORITY ACTIONS (queued)

Immediate (Priority 1-2 completion)

1. Re-run all WORKSTREAM-3 direct As-Ewing strings natively in PubMed/Europe PMC; log counts.
2. Same strings in OpenAlex + Semantic Scholar + Scholar; capture any grey/preclinical hits.
3. Full source records + sha256 for the 8 Priority-1 sources; promote to MASTER_BIBLIOGRAPHY.csv.
4. Dual-review Findings 1-3.
5. Pull IARC 100C arsenic full mechanistic section; extract genotoxicity/repair specifics.
6. WORKSTREAM 4A/B: arsenic + double-strand breaks / NHEJ / HR / repair inhibition - populate EWING_MECHANISM_MATRIX .
7. Search: arsenic + translocation-associated sarcoma; arsenic + topoisomerase; arsenic + chromothripsis.
8. Search: environmental/agricultural/pesticide risk factors for Ewing (was arsenic ever grouped under "pesticides"?).
9. NCI PDQ + key Ewing etiology reviews -> EWING_SARCOMA_KNOWN_UNKNOWN_MATRIX.
10. Cattle-dip chemistry (WORKSTREAM 6): arsenic trioxide vs sodium arsenite/arsenate; CAS crosswalk; vat-sludge/soil As speciation.

Near-term (Priorities 3-5)

- 11-18. Ewing developmental-window biology; germline loci; ancestry incidence; prenatal/parental-occupation studies.
- 19-24. Other reported cancers: per-cancer arsenic evidence class (osteosarcoma, leukemia, lymphoma, CNS, thyroid, etc.).
- 25-28. Biomarkers recoverable years pre-diagnosis: baby-tooth growth rings, archived newborn blood spots, toenail, FFPE-tumor As.
29. Molecular signatures: is there a validated arsenic mutational/methylation signature testable (blinded) on Ewing FFPE?

Governance / preservation

30. IMMEDIATE_EVIDENCE_PRESERVATION_CHECKLIST: which specimens/records could be lost (FFPE blocks, newborn screening cards, soil before any remediation/grading) - physician/attorney action list.

Doctor Technical Brief

DOCTOR TECHNICAL BRIEF (v0.1, 2026-07-23)

Physician-facing synthesis of the current evidence. **Preliminary, single-reviewer, hypothesis-neutral.** Sourced findings live in 05_/06_/07_/10_. Nothing here asserts causation.

Levels: a fact | b strong | c suggestive | d plausible | e hypothesis | g unknown.

1. What is known (a)

- Inorganic arsenic is an IARC Group 1 human carcinogen - with sufficient human evidence for bladder, lung, skin (epithelial), and an established hepatic angiosarcoma (a sarcoma).
- Ewing sarcoma is defined by a t(11;22) EWSR1::FLI1 (FET::ETS) fusion; cell of origin is a developmentally restricted mesenchymal stem/progenitor cell (embryonic-stage most permissive).
- Arsenic is genotoxic and suppresses homologous-recombination repair (not NHEJ), a state associated with chromosomal translocations and genomic instability.

2. What is suspected / plausible (c-d)

- A plausible but unproven mechanistic bridge: arsenic-driven demotion of high-fidelity repair + genomic instability could, in principle, favor formation or persistence of a translocation such as EWSR1::FLI1 - never tested at the EWSR1 loci or in Ewing cells of origin.
- Ewing has a **real, replicated epidemiological association with farm/agricultural/pesticide exposure** (maternal pesticide handling; paternal agricultural work, RR up to ~6). Historical arsenicals were agricultural pesticides - but those studies did not isolate arsenic.

3. What is not known (g)

- Whether environmentally relevant iAs increases EWSR1-region DSBs, mis-repair, or selects fusion-positive progenitors. Whether Ewing tumors carry any arsenic molecular signature. Whether any Ladera case had measurable historical arsenic exposure preceding diagnosis.

4. Evidence directly connecting arsenic to each reported cancer

See 06_other_cancers matrix. Summary: **bladder/lung/skin established; kidney/liver/prostate suggestive; hepatic angiosarcoma established; STS limited/occupational; osteosarcoma & Ewing - no direct evidence.**

5. Tumor tests that could strengthen or weaken the hypothesis (do not initiate - design for review)

- Blinded molecular analysis of Ewing FFPE: WGS/long-read (structural variants), RNA-seq, methylation array; metallomics (tumor arsenic); test against any validated arsenic mutational/methylation signature. State clearly when a signature is non-specific.

6. Exposure biomarkers still recoverable (see 10_ + preservation checklist)

- Deciduous teeth (weekly-resolution prenatal->early-childhood metals, LA-ICP-MS; As feasible), infant/adult toenails & hair (validated As biomarkers), archived newborn dried blood spots (state-banked), FFPE tumor arsenic. These can reconstruct exposure years before diagnosis - secure before disposal.

7. Studies physicians should design (priority)

1. Blinded Ewing tumor molecular + metallomic study (cases vs sporadic controls).
2. Biomarker exposure reconstruction (teeth/nails/blood-spots) in cases + matched comparisons.
3. Mechanistic: env-relevant iAs in embryonic/pediatric mesenchymal progenitors -> EWSR1 DSB/fusion/repair/selection assays (see 16_proposed_experiments).

—4. CDC/ATSDR-compliant cluster evaluation with verified (not social-media) case counts.

8. Specialists / labs needed

Pediatric oncologist (Ewing), molecular pathologist, sarcoma cytogeneticist, environmental toxicologist (arsenic speciation/bioavailability), exposure scientist (dental/nail biomarkers, LA-ICP-MS), environmental epidemiologist, biostatistician (rare-event/spatial), CA Cancer Registry.

9. What would falsify or materially weaken the hypothesis

- No measurable historical arsenic exposure in affected individuals (teeth/nails/blood-spots null).
- Soil at documented dip ground shows no low-lead arsenic-trioxide signature above background.
- Ewing tumors show no arsenic-associated signature and typical sporadic genomics.
- Cases dominated by known germline predisposition / prior radiotherapy.
- Rigorous cluster analysis shows incidence within expected range for the true denominator.

Any of these would not "disprove arsenic universally," but would remove this community's case for it.

Counter-evidence to weigh (07_): As₂O₃ treats the translocation leukemia APL; EWS-FLI1 alone does not transform normal cells (cooperating events needed); Ewing has a strong germline/ancestry component; arsenic's proven cancers are epithelial.

First Doctor Review Packet

FIRST DOCTOR REVIEW PACKET (v0.1, 2026-07-23)

A 5-minute orientation for the physician's first review. Preliminary, single-reviewer, hypothesis-neutral. Full detail in the linked files. Levels: a fact | b strong | c suggestive | d plausible | g unknown.

The question

Could historical inorganic arsenic (cattle-dip era) plausibly cause the reported Ewing sarcoma (and other cancers) in Ladera Ranch residents?

The 6 things to know first

1. No etiologic literature exists. Reproducible native-database result: arsenic AND EWSR1 = 0, arsenic AND EWS-FLI1 = 0 in PubMed. The only arsenic-Ewing papers (n=8) are **arsenic trioxide as a candidate therapy - it kills Ewing cells. (05_, 01_NATIVE_DB.) -> a**
2. Arsenic is a proven carcinogen, but of the wrong tissue type - bladder/lung/skin (epithelial), plus hepatic angiosarcoma. Not bone/mesenchymal. (06_) -> a
3. The mechanism arsenic is known for is the wrong mechanism. Its dominant MOA is cytotoxicity/regenerative proliferation (often non-genotoxic) - not making a translocation. A minority genotoxic thread exists (arsenite impairs HR/BRCA1-dependent DSB repair; PMID 37762697, 19553191) but has never been linked to EWSR1. (07_, 14_) -> d
4. Ewing's real environmental signal is "farm/pesticide/parental occupation," not arsenic - and arsenic was never isolated within it. Confounders: orchard lead-arsenate, DDT, ancestry. (12_, 13_) -> c
5. Local exposure has never been measured. Analog dip sites hold up to ~3,000 mg/kg arsenic decades on (low-lead "dip signature"), but no local soil/biomarker/tumor data exist. (08_, 14_) -> d/g
6. Exposure is still reconstructable - but degrading. Baby teeth, toenails, archived newborn blood-spot cards, and FFPE blocks can reconstruct exposure years pre-diagnosis; soil signature is lost once ground is disturbed. Preserve now. (10_, 20_PRESERVATION) -> a/b

Net

Causation is currently unsupported/unestablished. The environmental predicate is **plausible but unmeasured. The defensible path is measurement**, not assertion.

What the physician can do that would move the evidence (design only; GATE B authorizes)

- Confirm each case's pathology + EWSR1 fusion status/partner; assemble residence/school and prenatal/parental windows privately (04_).
- Support a blinded tumor protocol (metallomics + methylation + EWSR1 breakpoint; low-mutation-burden caveat, 09_) and a biomarker reconstruction (teeth/nails/blood spots, 10_).
- Back legal holds on blood-spot cards and FFPE blocks, and authorized soil speciation before any ground disturbance.

What would falsify/weaken (so the review is balanced)

Null soil dip-signature + null reconstructed exposure would substantially weaken the community-specific case even before mechanism work. Cases dominated by germline predisposition/prior radiotherapy, or an ancestry-standardized incidence within expected range, would also weaken it. (20_TOP_15_FALSIFIERS.)

Key linked files

05_ direct findings | 06_ other cancers | 07_ mechanistic bridge | 08_ dip chemistry |
09_ molecular-signature protocol | 10_ epidemiology plan | 12_ causal weight-of-evidence |
13_ alternatives | 14_ proposed experiments | 20_ preservation checklist + gaps/hypotheses/falsifiers |

22_ attorney brief.

Attorney Science Brief

ATTORNEY SCIENCE BRIEF (v0.1, 2026-07-23)

Scientific, not argumentative. **Preliminary, single-reviewer, hypothesis-neutral. No legal conclusions.** Distinguishes general vs specific causation. Levels: a fact | b strong | c suggestive | d plausible | e hypothesis | g unknown.

1. Which links in the causal chain have evidence, and which are missing

Chain: **i) arsenic used at the parcel -> ii) arsenic released/retained in soil -> iii) soil redistributed to child-contact areas -> iv) residents exposed -> v) exposure preceded diagnosis -> vi) arsenic capable of causing the specific cancer -> vii) it did in these individuals.**

Link | Status | Note

- i Arsenic used at the parcel | g / needs location-specific evidence | statewide mandate != parcel proof; 1908 Joplin dip documented on Trabuco/Bell Canyon ground (historical investigation)
- ii Release/retention in soil | d (arsenic persists; element, no decay) | not yet measured at these parcels
- iii Redistribution by grading | e/g | plausible via mass grading; unmeasured
- iv Resident exposure | g | no biomarker data yet
- v Exposure preceded diagnosis | g | reconstructable via teeth/nails/blood-spots (10_)
- vi General causation, As -> this cancer | bladder/lung/skin a; hepatic angiosarcoma a; Ewing g | arsenic can cause a sarcoma; no evidence for Ewing/bone
- vii Specific causation | g | not established

2. Which assertions are presently too strong (do not advance as fact)

- "Arsenic causes Ewing sarcoma." - unsupported (no direct study).
- "Dip vats existed within the development footprint." - needs location-specific evidence (Ladera footprint: none documented; Bell Canyon/Coto: 1908 dip documented nearby).
- "Contaminated soil was redistributed and residents were exposed." - unverified (no soil/biomarker data).
- "Cancer incidence is statistically elevated." - requires CDC/ATSDR analysis with verified counts (not social-media tallies). (See 12_epidemiology.)

3. General vs specific causation (Daubert-relevant)

- General causation (can arsenic cause the disease type): **established for bladder/lung/skin and hepatic angiosarcoma; NOT established for Ewing/bone sarcoma** - currently a research gap, not proof either way.
- Specific causation (did it cause these plaintiffs' cancers): unestablished; would require documented parcel arsenic, reconstructed individual exposure preceding the etiologic window, and a scientifically accepted mechanism/biomarker linking the two.

4. Records/datasets that would resolve major uncertainties

Location-specific: ranch ledgers/receipts, County Vet orders, State Vet files (archive list).
Environmental: authorized soil sampling (total+bioaccessible As, lead, speciation) at priority grounds. Medical: pathology + EWSR1 status, residence/school histories (private). Denominators: CA Cancer Registry, SEER.

5. Evidence requiring formal chain of custody

Soil samples; biological specimens (teeth, nails, blood-spot cards, FFPE blocks); original archival documents. Sampling should occur before any soil disturbance/remediation (link ii/iii degrade).

6. Evidence that may disappear with time (see IMMEDIATE_EVIDENCE_PRESERVATION_CHECKLIST)

Baby teeth, nails, archived newborn blood-spot cards (retention schedules), FFPE blocks (hospital limits), predevelopment soil/boring archives, private ranch records. **Legal holds are time-sensitive.**

7. Alternative causes / confounders that must be addressed (15_)

Lead-arsenate (orchard) vs arsenic-trioxide (dip) signatures; organochlorines (DDT/chlordane); other metals; germline Ewing predisposition; ancestry; chance aggregation; ascertainment/ media-recruitment bias; residential mobility; other environmental sources.

8. Potential experts by issue

Molecular pathology/cytogenetics; arsenic toxicology + speciation; exposure science (dental/nail biomarkers); environmental epidemiology; biostatistics (rare events/spatial); geomorphology/soil (redistribution); pediatric oncology.

9. Bottom line for counsel (scientific)

The historical/ environmental predicate is partially documented; the medical-causation predicate for Ewing is currently unproven and largely unstudied. The defensible posture is to preserve evidence and commission the specific studies (soil signature, individual biomarker reconstruction, blinded tumor molecular analysis, compliant cluster evaluation) that would move each link from plausible/unknown toward established or excluded - not to assert causation now.

Draft Preservation Letters

DRAFT PRESERVATION & RECORDS-REQUEST TEMPLATES (for the legal team; NOT to be sent by this engine)

Status: DRAFTS for the supervising attorney to review, adapt, and - at their discretion - send.

This engine does not contact any person, agency, custodian, or family, and draws no legal conclusions. Bracketed [...] fields are placeholders. Purpose: stop loss of time-sensitive evidence (Phase 3), pending GATE B. Nothing herein is legal advice.

Template A - Litigation/evidence hold: archived newborn dried blood-spot cards

To: [State newborn-screening program / biobank custodian]

Re: Preservation request - dried blood-spot specimens

"[Firm/Physician] requests that the [program] preserve and not destroy or deaccession the dried blood-spot specimens and associated records for [individuals/cohort - identified separately under appropriate authorization], which may be material to an ongoing health investigation. We ask that routine destruction schedules be suspended for these specimens pending further request, and that you confirm current retention status and the process for authorized research access. [Consent/authority basis]. Please preserve chain of custody."

Template B - Evidence hold: archived FFPE tumor blocks & pathology materials

To: [Hospital/pathology department custodian]

Re: Preservation of pathology specimens

"Please preserve and do not exhaust, discard, or return the formalin-fixed paraffin-embedded blocks, slides, and associated pathology records for [patient(s), per authorization], including any remaining tissue, pending authorized research/legal analysis. Kindly confirm retention policy and remaining material. Maintain chain of custody."

Template C - Family keepsake retention (routed via the family's own clinician/counsel)

Note for the team: do not contact families directly through this project. Via their own clinicians/counsel, provide guidance to retain shed baby teeth, nail clippings, and similar keepsakes (do not discard), as these may enable exposure reconstruction. No collection occurs without IRB approval + consent.

Template D - Records request: ranch/county/state archives (historical exposure geography)

To: [Sherman Library / Orange County Archives / California State Archives / private RMV archive]

Re: Access & preservation - cattle-dip-era ranch and veterinary records (1900-1920)

"We request access to, and preservation of, records that may document cattle-dipping operations and vat locations, including: ranch business ledgers, foreman correspondence/diaries, cattle accounts and dipping/arsenic-purchase receipts; County Veterinarian orders and Board of Supervisors minutes; State Veterinarian biennial reports and Cattle Protection Board files (record groups [F3741/F3742/F3744]); and ranch photograph collections. Please advise on access procedures and confirm these materials are retained pending review."

Template E - Agency/landowner permission: authorized soil sampling

To: [Landowner / National Audubon-Starr Ranch / relevant agency]

Re: Request for authorized, chain-of-custody soil sampling

"We request permission to conduct authorized, non-destructive-as-feasible soil sampling at [location] under professional chain of custody, to characterize legacy soil chemistry (arsenic speciation and related analytes). Sampling would follow the attached Sampling & Analysis Plan and be

performed by [qualified firm]. We will share results and coordinate on timing ****before any ground**
disturbance/remediation. ****"**

Custodian / hold tracker (to be maintained by the legal team)

Item | Custodian | Hold requested (date) | Confirmed | Access path | Degradation risk

Newborn blood-spot cards | [state program] | retention schedule

FFPE blocks | [hospital] | exhaustion/discard

Baby teeth/nails | family (via counsel) | disposal

Ranch/county/state archives | [repositories] | deaccession

Priority soil | [landowner/agency] | grading/remediation

DRAFTS ONLY - for attorney review and action. This engine sends nothing and contacts no one.

Appendix - Data Files (in repository)

Structured datasets accompanying this report in the repo (arsenic-cancer-investigation/). Row counts shown.

- 02_source_library/TOP_100_MOST_RELEVANT_SOURCES.csv (118 rows)
- 17_expert_directory/EXPERT_DIRECTORY.csv (10 rows)
- 18_evidence_matrices/EWING_MECHANISM_MATRIX.csv (5 rows)
- 18_evidence_matrices/EWING_SARCOMA_KNOWN_UNKNOWN_MATRIX.csv (5 rows)
- 18_evidence_matrices/EXPOSURE_BIOMARKER_MATRIX.csv (5 rows)
- 18_evidence_matrices/MASTER_BIBLIOGRAPHY.csv (34 rows)
- 18_evidence_matrices/OTHER_CANCERS_ARSENIC_MATRIX.csv (11 rows)
- 18_evidence_matrices/STUDY_EVIDENCE_MATRIX.csv (0 rows)
- 19_claims_audit/CLAIMS_AUDIT.csv (17 rows)